



Stanford | Environmental
Health & Safety

SCHOOL of MEDICINE LAB SAFETY COORDINATOR BINDER

Laboratory Safety Coordinator Binder

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Assignment of Laboratory Safety Tasks to Laboratory Safety Coordinators

Per Stanford University's Research Policy Handbook, the Principal Investigator/Laboratory Supervisor is responsible for the health and safety of laboratory personnel doing work in his/her laboratory. The PI/Lab Supervisor may delegate certain safety tasks to one or more lab members. These lab members are often referred to as Laboratory Safety Coordinators (LSCs), but may also be called Lab Safety Contacts or Lab Managers. The PI/Lab Supervisor still retains ultimate responsibility and must make sure that any delegated tasks are carried out.

Instructions to the PI/Lab Supervisor

Assigning Health and Safety Tasks: If you elect to delegate routine health and safety tasks to one or more members of your lab, use this checklist or an equivalent form to document task assignments. You may use the "Other Duties" section to assign other safety duties you deem appropriate. If, at any point, a new laboratory safety coordinator is appointed, update this form accordingly. EH&S recommends that outgoing LSCs overlap with their replacement for an extended period of time (e.g. 1-3 months) to ensure adequate knowledge transfer.

Completing Health and Safety Tasks: Refer to the following pages for guidance on Commonly Delegated Health & Safety Tasks. Stanford EH&S can provide more in-depth training to LSCs. Contact your Research Safety Specialist or the EH&S office (650) 723-0448 to sign up for Orientation for Laboratory Safety Coordinators (EHS-5200).

Lab Name: _____

Building/Rooms: _____

Task	Name of Assignee(s)	Date Assigned
A. Safety Training		
B. Standard Operating Procedures (SOPs) / Risk Assessment		
C. Laboratory Self-Inspections		
D. Chemical Inventory & Management		
1. Chemical Storage, Segregation, and Labeling		
2. Chemical Waste Management		
3. Life Safety Box Updates		
E. Incident Reporting		
F. Additional Safety Support		
1. EH&S Lab Visits		
2. Lab Ergonomics		
3. Biosafety (if applicable)		
4. Radiation Safety (if applicable)		
G. External Regulatory Inspections		
H. Other Duties:		

X _____
PI/Lab Supervisor Signature

Date

Commonly Delegated Health & Safety Tasks

A. Safety Training

1. Train New Lab Members
 - a. **Tier I** Training is required for all personnel - General Safety & Emergency Preparedness - Available on-line in [AXESS/STARS](#). (EHS-4200)
 - b. **Tier II** Training is determined based upon your work responsibilities using the [Training Needs Assessment](#) on-line in [AXESS/STARS](#). Training will be assigned in STARS based upon the assessment.
 - c. **Tier III** is [Laboratory-specific training](#) for new lab members prior to conducting laboratory work. Include the [Personal Protective Equipment \(PPE\) Assessment Tool](#).
 - i. Recordkeeping - maintain all Tier III and other in-person training records for a minimum of one year (unless otherwise specified). Records may be stored in [BioRAFT](#) under the *Document* tab or as hard copies.
2. Develop and/or update Laboratory-specific training
 - a. Work with PI/Lab Supervisor to establish and maintain a system by which lab members can obtain additional training prior to new procedures involving new potential exposure situations and/or hazardous operations.
 - b. Assist PI/Lab Supervisor in development and revision of PI's Lab-Specific training program using the following templates:
 - i. [How to Develop Lab-Specific Training](#) or [Laboratory-Specific Checklist](#)
 - ii. [Personal Protective Equipment \(PPE\) Assessment Tool](#)

B. Standard Operating Procedures (SOPs) / Risk Assessment

1. Work with PI/Lab Supervisor to ensure that the lab has appropriate SOPs in place.
 - a. Lab members must have an SOP approved by the PI/Lab Supervisor prior to working with [SU Restricted Chemicals](#) (e.g., [toxic gases](#) regulated by Santa Clara County, dimethylmercury, [hydrofluoric acid](#), *tert*-butyllithium, and DoT 1.1 explosives).
 - b. Lab members must consult with PI/Lab Supervisor to prioritize development of SOPs for work involving highly toxic chemicals, carcinogens, reproductive toxins, highly reactive materials, and other high risk operations. Refer to [General Use SOP for Highly Reactive/Unstable Materials](#).
 - c. Additional [SOP Guidance](#) include [templates](#) and General Use SOPs. EH&S is available to perform courtesy reviews of SOPs. Please use this form to submit your lab-specific SOPs if you would like EH&S to review them, or you are working with chemicals or procedures for which an EH&S review is required:
<https://app.smartsheet.com/b/form/6af14bd85ac64e2b90ddd7f16ee91fdb>
 - d. [Safety Fact Sheets](#) for commonly used materials and procedures can serve as starting points and references for laboratory personnel.
2. Evaluate the risk of an experiment, analytical process or task, researchers can use [Stanford Laboratory Risk Assessment Tool](#). Contact EH&S at (650) 723-0448 with any questions or to request support in conducting a risk assessment.

C. Laboratory Self-Inspections

1. Ensure [laboratory self-inspections](#) are completed using [BioRAFT](https://stanford.bioraft.com) (<https://stanford.bioraft.com>) or the templates linked below:
 - a. Required quarterly self-inspections for [lab areas](#) (Jan-Mar, Apr-Jun, July-Sept, Oct-Dec)
 - b. Required monthly self-inspections for [hazardous materials storage areas](#) (isolated areas not routinely occupied by lab personnel. Contact EH&S if you are unsure if a room requires monthly self-inspections)
 - c. If applicable, [quarterly self-inspections](#) for Controlled Substances (recommended)
 - d. If applicable, [annual self-inspection](#) for Laser Safety
2. Assist PI/Lab Supervisor in correcting any deficiencies identified during self-inspections, documenting corrective actions in BioRAFT or on self-inspection sheets, and reviewing findings from self-inspections during group meetings and via other internal lab communication systems.
3. Refer to the [Lab Compliance Cheat Sheet](#) for violations commonly cited by the County of Santa Clara during hazardous materials inspections.
4. Maintain records for a minimum of **three** years.

D. Chemical Inventory and Management

- Ensure chemical inventory is maintained on an on-going basis using [ChemTracker](#). See the [Chemical Inventory Management](#) page for a list of materials that must be inventoried.
- Familiarize lab members with the ChemTracker application and inventorying requirements. Contact your [Research Safety Specialist](#) to arrange individual or lab-wide training.
 - Go to <http://chemtrackerhelp.stanford.edu> for account-related assistance, including creating new user accounts and resetting passwords for existing accounts.

1. Chemical Storage, Segregation, and Labeling

- a. Periodically check lab practices regarding chemical storage, segregation, and labeling.
 - i. Ensure flammable materials are stored in an appropriate flammables cabinet.
 - ii. Ensure that all hazardous chemicals are stored in appropriate secondary containment. Refer to the guidance on [Primary and Secondary Container Criteria](#).
 - iii. Ensure chemicals are stored compatibly using the [Stanford Storage Group system](#). Storage Groups for specific chemicals can be determined using the [Stanford Chemical Safety Database](#) (also accessible through ChemTracker).
 - iv. Ensure all containers are clean, free of cracks and other structural defects, and labeled with their full English name; chemical abbreviations and formulas are not sufficient.
 - v. Assist PI/Lab Supervisor in discussing findings during group meetings and via other internal lab communication systems to ensure lab-wide engagement and follow-through.

2. Chemical Waste Management

- a. Periodically check lab practices regarding labeling and [management of chemical waste](#).
 - i. Containers must be labeled with a waste tag before any waste is deposited.
 - ii. Waste tags can be created and managed via <http://wastetag.stanford.edu>
 - iii. Wastes must be stored compatibly and in secondary containment.
 - iv. EH&S maintains a list of [non-hazardous chemical wastes](#) and recommended methods for their disposal.

- v. Assist PI/Lab Supervisor in discussing findings during group meetings and via other internal lab communication systems to ensure lab-wide engagement and follow-through.
- b. Arrange for pickup of hazardous waste containers after 8 months of waste accumulation or when the container is full, whichever occurs first.
 - i. Pickup of containers labeled with barcoded waste tags must be requested through <http://wastetag.stanford.edu>
 - ii. Pickup of containers labeled with non-barcoded waste tags must be requested through <http://wastepickup.stanford.edu>

3. Life Safety Box Updates

- a. Ensure contents of the Life Safety Box (emergency contact information and chemical storage map of laboratory) are current to within the last 12 months.
 - i. [Blank Emergency Contact Sheet \(University Main Campus\)](#)
 - ii. [Blank Emergency Contact Sheet \(School of Medicine\)](#)
 - iii. [Blank Chemical Storage Map](#)
- b. Request new ChemTracker inventory printout and cover sheet: If over a year old or significant changes to inventory: Go to <http://chemtrackerhelp.stanford.edu>

E. Incident Reporting

- 1. Be familiar with the [Serious Injury/Illness Reporting Procedures](#)
- 2. In the event of lab incidents, injuries, or near-misses, assist PI/Lab Supervisor and involved parties in the completion and submission of an Incident Investigation Report (eSU-17 form), available at <https://ehs.stanford.edu/esu-17>
- 3. Assist PI/Lab Supervisor in correcting conditions that were causal factors of the incident and documenting such efforts. Retrain lab personnel as appropriate.
- 4. Assist PI/Lab Supervisor in reviewing incidents and near-misses during group meetings and via other internal lab communication systems. This may involve:
 - a. Reviewing how to minimize reoccurrence
 - b. Reinforcement of correct use of engineering controls, work practices, and personal protective equipment (as appropriate)
 - c. Promoting reporting of near misses
 - d. Reviewing the University's Anti-Reprisal Policy, which allows for reporting of safety concerns without fear of negative repercussions.

F. Additional Safety Support

1. EH&S Lab Visits

- a. Participate in laboratory visits with EH&S staff on topics such as:
 - i. General lab safety (e.g., SOP development, chemical inventory, hazardous materials storage, and waste management)
 - ii. Fire safety
 - iii. Biosafety (if applicable)
 - iv. Health Physics (if applicable)
 - v. Controlled Substances (if applicable)

- b. Address issues on the spot, when feasible (e.g., completion of hazardous waste tags, placing hazardous materials in secondary containment).
- c. Assist PI/Lab Supervisor with implementing any corrective actions or best practice suggestions identified by EH&S representatives.

2. Lab Ergonomics

- **Laboratory Ergonomics Training – EHS 4800:** Prior to beginning work in a lab, EHS recommends completion of Lab Ergonomics Training in Axess/STARS.
- **Laboratory workstation evaluations:** For assistance with your laboratory workstation setup and ergonomic products, please contact EH&S Ergonomics team [here](#).
- **Laboratory ergonomics webpage:** <https://ehs.stanford.edu/subtopic/laboratory-ergonomics> for the following topics:

Lab Ergo Self Evaluation Form	Laboratory Ergonomics Catalog
Lab Ergo Matching Fund Program	Ergo Tip Sheet Fume Hood
Ergo Tip Sheet Microscopy	Ergo Tip Sheet Pipetting
Adjusting Lab Stools	Microbreaks

3. Biosafety (if applicable)

- a. Ensure biosafety cabinets (tissue culture hoods) are [certified annually](#).
- b. Arrange for pickup of [biohazardous waste](#). For assistance, call (650) 724-0794 (University Main Campus) or (650) 721-2146 (School of Medicine).

If the lab uses human or non-human primate cells (including tissue culture), blood, blood products, body fluids, organs, or other potentially infectious materials:

- Assist PI/Lab Supervisor in ensuring researcher compliance with [Universal Precautions](#). Refer to the [Biosafety Manual](#) or contact the [Biosafety program](#) for more information.

If the lab uses biologically hazardous agents (Biosafety levels 2 and higher) and/or non-exempt recombinant or synthetic nucleic acids:

- Assist PI/Lab Supervisor with submitting, revising, and renewing APB protocols through <http://eprotocol.stanford.edu>

4. Radiation Safety (if applicable)

If the lab uses/will use radioactive material or any machine that emits ionizing or non-ionizing radiation (including lasers):

- Visit the [radiation safety](#) and [laser safety](#) websites for a description of relevant safety duties.
- Contact Health Physics at (650) 723-3201 for additional assistance.

G. External Regulatory Inspections

- Whenever possible, be present during external regulatory inspections to address questions posed by inspectors regarding laboratory operations, including hazardous materials storage/waste management and review of quarterly and monthly lab self-inspection records.
 - EH&S staff (and/or Departmental Health & Safety Programs staff) will escort external regulatory inspectors, along with the local area contact.
- Correct violations on the spot during the inspection, when feasible (e.g., completion of a hazardous waste tag) and as allowed.

Stanford University
Laboratory Safety Coordinator Guidance

- Assist PI/Lab Supervisor with: (1) correcting outstanding violations and (2) providing the required responses documenting the abatement of the violations to the local area contact within the required time frame.

H. Other Duties

Other lab-specific safety duties may be assigned as the PI/Lab Supervisor deems appropriate.

Resources

EH&S	ehs.stanford.edu
Emergency Preparedness Checklist for students, faculty, staff, and visitors	ehs.stanford.edu/wp-content/uploads/EmerPrepChecklist.pdf
Laboratory Continuity Planning	ehs.stanford.edu/topic/emergency-preparedness/continuity-planning
Laboratory Ergonomics	ehs.stanford.edu/wp-content/uploads/lab_ergo_tips.pdf ehs.stanford.edu/wp-content/uploads/Lab-Ergo-Product-List.pdf
BioRAFT	stanford.bioraft.com
Chemical Inventory	stanford.chemtracker.org
Chemical Safety Database	chemsafetydata.stanford.edu
Chemical Toolkit	chemtoolkit.stanford.edu
Free Chemicals	freechemicals.stanford.edu
Forms	ehsforms.stanford.edu
Hazmat Shipping	hazmatshipping.stanford.edu
Online Safety Store	ehs.stanford.edu/safety-store
Online Training	axess.stanford.edu
Radiation	radsafety.stanford.edu
Waste Pickups	wastepickup.stanford.edu
Waste Tag	wastetag.stanford.edu

SAFETY TRAINING

Safety Training

<https://ehs.stanford.edu/topic/chemical-safety/safety-training>

Information and training must be made available to educate laboratory personnel of the hazards of chemicals present in their work area.

For work directed by the PI or laboratory supervisor

Laboratory personnel must receive general and laboratory-specific information and training when initially assigned to the laboratory, and prior to assignments involving new exposure situations, particularly hazardous substances, and hazardous operations.

Safety Training

What to do?	How to do it?
Obtain general laboratory safety training, which includes orientation to and training on the Chemical Hygiene Plan. All classes are available online in STARS at http://axess.stanford.edu/	Take the following training. • General Safety & Emergency Preparedness (EHS-4200) • Chemical Safety for Laboratories (EHS-1900)* And, where applicable: • Compressed Gas (EHS-2200)* • Biosafety (EHS-1500)* • Computer Workstation Ergonomics (EHS-3400) • Laboratory Ergonomics (EHS-4800) * School of Medicine ONLY - Life Sciences Research Laboratory Safety Training (EHS-4875). EHS-4875 is inclusive of Biosafety (EHS-1500), Chemical Safety in Laboratories (EHS-1900), and Compressed Gases Safety (EHS-2200). Researchers not in a SOM laboratory, and not in a Life Sciences Research laboratory should instead take the separate courses as applicable.
Obtain laboratory-specific training	1. See your PI or laboratory supervisor to review the lab-specific training procedures for your lab. 2. Review any individual Laboratory Safety Plan. 3. Complete lab-specific training* or equivalent, which includes: • Local/building safety information • Standard Operation Procedure(s) involving hazardous materials (you may use the Documenting SOP & PI Approval to document your review) • Any other laboratory-specific safety procedures or hazards that may be encountered in the laboratory environment (lab-owned equipment may require specialized training to ensure safety and prevent equipment damage) * Additional guidance for PIs on how to develop lab-specific training can be found in How to Develop Lab-Specific Training.
Maintain Training Records	1. The PI or laboratory supervisor, or designate, must retain training documents for lab personnel for at least one year. 2. Use How to Develop Lab-Specific Training or equivalent to document training.

Hazard information

The following hazard information is available for PIs, lab supervisors, and lab personnel to consult during the experiment planning process to assess the hazards and potential risks associated with chemicals and lab operations.

For information on:	See:
Reference materials on the hazards, signs and symptoms of exposure, safe handling, storage, and disposal of hazardous chemicals	• Safety Data Sheets • Stanford University's Chemical Safety Database • Chemical Safety, National Institute of Health • SOPs
Cal/OSHA's Permissible Exposure Limits	"Permissible Exposure Limits (PEL) for Chemical Contaminants." California Code of Regulations, Title 8, Section 5155 Cal/OSHA establishes regulatory exposure limits for many airborne contaminants. The actual values are in Table AC. If a PEL is not established for a specific contaminant, contact EH&S for guidance.
Cal/OSHA's Laboratory Standard	"Occupational Exposure to Hazardous Chemicals in Laboratories." California Code of Regulations, Title 8, Section 5191 Cal/OSHA is a governmental agency that protects worker health and safety in the State of California. This regulation was promulgated to protect laboratory personnel engaged in the laboratory use of hazardous chemicals. (Note that custodial and maintenance staff who service the laboratory fall under Cal/OSHA's Hazard Communication Standard, Code of Regulations Title 8, Section 5194.
Stanford University's Chemical Hygiene Plan	Stanford University's Chemical Hygiene Plan The above-referenced Cal/OSHA regulation requires employers to have a written Chemical Hygiene Plan. This Plan fulfills this regulatory requirement and is a resource for planning experiments and laboratory operations.

For work conducted autonomously or independently

What to do?	How to do this?
Consult with the PI or laboratory supervisor	1. The PI or laboratory supervisor (or designate) must retain training documents for lab personnel for at least one year. Note that training records for EH&S-provided training are maintained electronically. 2. Obtain any other training that is appropriate to your work in Stanford University labs. • Provide appropriate oversight, training, and safety information to any lab personnel you supervise or direct.

REQUIRED TRAINING FOR LAB RESEARCHERS & PIs

These courses in STARS axess.stanford.edu are designed to ensure compliance with applicable external regulatory requirements and Stanford University guidelines and policies.

Course title and STARS number:	Must be taken by all who:	Course Notes:
General Safety, Injury Prevention (IIPP) and Emergency Preparedness EHS-4200-WEB	...work at Stanford University	
Life Sciences Research Lab Safety EHS-PROG-4875	...work in life sciences research laboratories in the School of Medicine with biological agents, hazardous chemicals & compressed gases*	
Biosafety EHS-1500-WEB	...work with biological agents*	
Chemical Safety for Laboratories EHS-1900-WEB	...work with chemicals*	
Compressed Gas Safety EHS-2200-WEB	...work with compressed gas cylinders*	
Laboratory Ergonomics EHS-4800	...perform repetitive tasks such as microscope use, pipetting, and miscellaneous hand tool use	
Computer Workstation Ergonomics EHS-3400-WEB	...use a computer routinely	
Bloodborne Pathogens EHS-PROG-1600 EHS-PROG-1601 (Recert)	...work with human and/or non-human primate blood, blood products, cells (including tissue culture) or other potentially infectious material	Must be taken annually.
Aerosol Transmissible Diseases EHS-PROG-1090 EHS-PROG-1091 (Recert)	...work with aerosol transmissible pathogens	Must be taken annually and review the Lab ATD Biosafety Plan annually.
Radiation Safety Training EHS-5250-WEB	...have never worked with radioactive materials before	Must also take EHS-5251 Hands-on Training.
Working Safely Near Radioactive Materials EHS-5275	...do not use radioactive materials, but work nearby	

Laser Safety EHS-PROG-4820 EHS-PROG-4821 (Recert)	...work with Class 3 or 4 lasers	Must be taken every three years.
DOT: Excepted Quantities EHS-PROG-2650 EHS-PROG-2651 (Recert)	...who are required to identify, package and air ship small quantities of hazardous chemical materials	Must be taken every two years.
DOT: Shipping Dangerous Biological Goods or Dry Ice EHS-PROG-2700 EHS-PROG-2701 (Recert)	...package and/or ship dangerous biological materials, or who package and/or ship any packages containing DRY ICE	Must be taken every two years.
Cryogenic and Dry Ice Safety EHS-2480	...work with cryogenic liquids and/or dry ice	Request course at email address below.
Orientation for Lab Safety Coordinators EHS-5200	... are assigned safety/compliance tasks by their PI or lab supervisor	Request course at email address below.
Controlled Substances EHS-2125-WEB	...work with controlled substances	
Fire Extinguisher Training EHS-3700 (classroom) EHS-3850-WEB	...work at Stanford University	
Back Care/Safe Lifting and Carrying EHS-1400-WEB	...perform material handling tasks or with back health concerns.	
Electrical Safety Awareness EHS-2800	...use and care for electrical equipment, power strips, & are involved with electrical emergencies	
Haz Mat Response Cleanup Labs/Shops EHS-4400	... clean up small spills.	Request course at email address below.
Portable Ladder Safety for Researchers EHS-4830	...use portable ladders and to prevent falls. Take EHS-4810 for fixed ladders and roof access ladders.	
Other courses as necessary	This is not a complete list of safety training courses that you may be required to take	Please go to Training Needs Assessment Tool in STARS/Axess to identify the safety training that is required for your work.

* EHS-PROG-4875 contains all content in the three courses EHS-1500-WEB, EHS-1900-WEB, and EHS-2200-WEB. Take EHS-PROG-4875 if you work in a School of Medicine research laboratory. If you work in a School of Medicine research laboratory BUT work with only one or two of the Biological, Chemical or Compressed Gas categories, you should *instead* take the separate courses.

For questions, email: ehs-training@stanford.edu

STANFORD UNIVERSITY LABORATORY-SPECIFIC TRAINING CHECKLIST

Laboratory Name: _____ Building/Room(s): _____

- ☐ Add researcher to the "Laboratory Members" list in BioRAFT (Go to <https://stanford.bioraft.com/>)
- ☐ Show researcher how to access the STARS training (Go to <https://axess.sahr.stanford.edu/>) and assist them in completing a [Training Needs Assessment](#) for their research. Identify any additional training requirements.
- ☐ If applicable, add researcher to relevant [APB](#), APLAC and/or IRB protocols in the [eProtocol](#) program.

Review the location and proper use of the following safety features:

- ☐ Information in Life Safety Boxes (including emergency contacts).
- ☐ Emergency evacuation routes and [emergency assembly points](#).
- ☐ Fire extinguishers and closest fire alarm pull station.
- ☐ Safety showers and eyewash stations.
- ☐ Emergency shutoffs for laboratory equipment.
- ☐ First aid kits.
- ☐ Laboratory small spill kits and building spill kits or both.
- ☐ Chemical [fume hoods](#), [biosafety cabinets](#), and glove boxes.
- ☐ [SU-17 incident report forms](#).
- ☐ If applicable, specialized medical supplies (e.g. calcium gluconate for [hydrofluoric acid](#) and amyl nitrite for [cyanides](#)).

Review the following lab hazards, safety plans, procedures, and safety manuals:

- ☐ Location of highly hazardous materials, equipment, and processes present in the work area.
- ☐ Lab's [PPE assessment](#) and location, use and limitations of required PPE (e.g. gloves, safety glasses, lab coats, etc.)
- ☐ Applicable protocols, risk assessments, [Standard Operating Procedure \(SOPs\)](#), lab's local plans (e.g. BBP Exposure Control Plan), safety fact sheets, pathogen safety data sheets and/or work plans for hazardous materials, equipment, or processes present in your work area.
- ☐ Laboratory waste management (e.g. [chemical](#), [biological](#), [sharps](#), [radiological](#)).
- ☐ Location of Safety Data Sheets in lab or [online](#).
- ☐ Location of [Chemical Hygiene Plan](#), [Biosafety Manual](#), and [Radiation Safety Manual](#) in lab or online.
- ☐ If work will involve radioactive materials and/or lasers, contact EH&S Health Physics at 723-3202 to discuss training requirements.

☐ ***I understand that additional risk assessment and training¹ may be required when there is a change in the hazards associated with my work.***

By signing this form, I agree that I have been trained on the above checked items.

Trainee name (print): _____

Trainee signature: _____ Date: _____

Training was provided on the above checked items by:

PI/designee name (print): _____

PI/designee signature: _____ Date: _____

¹ Use SU's [Additional Laboratory-Specific or Job-Specific Training Log](#) to document subsequent lab-specific trainings for specific lab equipment, SOPs, etc.



HOW TO DEVELOP LAB-SPECIFIC TRAINING

SUMMARY

Stanford's Chemical Hygiene Plan requires that all lab members be trained on the specific hazards that exist in their lab and the procedures, equipment, and resources available in their lab for working safely with these hazards.

Lab-specific training must be: (1) documented and (2) provided to all lab personnel at the time of initial assignment to the lab and prior to work involving new exposure situations and hazardous operations.

HOW TO USE THIS TEMPLATE

The template below may be used to develop a lab-specific training handout and for documenting the training. After reviewing the training goals in the left-hand column, describe in the right-hand column how your lab fulfills these goals. Guidance text provided in **gray** should be modified and adapted to reflect your lab's practices. The guidance text may be deleted.

	TRAINING GOAL	HOW LAB FULFILLS TRAINING GOAL
TRAINING	Ensure completion of all safety training before beginning lab work.	Identify required training for each new lab member. At minimum should include completion of: - General Safety and Emergency Preparedness (EHS-4200; online via Axess) - Chemical Safety for Laboratories (EHS-1900; online via Axess) - Lab-Specific Training
SAFETY ROLES	Know the health and safety responsibilities of the principal investigator, lab safety coordinator, and all lab group members.	Describe the process for discussing and addressing health and safety concerns in the lab. Include information on expectations for all lab members. Identify additional key personnel for the building and/or department such as the facilities manager, Safety Chair, EH&S, Human Resources, etc.
SAFETY INFO	Know where to find safety data sheets (SDS), standard operating procedures (SOP), user manuals for equipment, journals, textbooks, etc.	Identify and list how to locate relevant resources for safety information including: - Chemical Hygiene Plan (http://chemhygieneplan.stanford.edu) and Cal/OSHA regulations - Resources specific to the lab such as a lab-specific SDS binder (if any), SOPs, etc.
LAB-SPECIFIC HAZARDS	Know the specific hazards that exist in the lab and which hazards are covered by existing SOPs.	In this section provide a basic overview of hazards present in the lab and any controls or alarms that all lab members must be aware of. For example, presence of lasers, biohazards, reproductive hazards, reactives, toxic gas, etc.

LAB OPERATIONS	Know the Chemical Hygiene Plan SOP requirements and the lab's process for developing and reviewing new SOPs.	Per SU Chemical Hygiene Plan the following materials receive priority for SOP development: highly toxic chemicals, carcinogens, reproductive toxins, and highly reactive materials. In addition, this section should review the PI's expectations for when written SOP development is triggered.
	Know the lab's chemical ordering, usage, and disposal procedures.	Include: Where lab chemicals are stored (including flammable cabinet locations), how ChemTracker is used for inventorying chemicals, and where hazardous waste is collected and what are the waste labeling procedures specific to the lab.
	Know what is required personal protective equipment (PPE) for working in the lab, including where lab-provided PPE is stored such as safety glasses/goggles, cryogenic gloves, etc.	See the Personal Protective Equipment section of the Chemical Safety Toolkit (chemtoolkit.stanford.edu) for minimum requirements. If PPE is not required at all times in the lab then identify the areas, times, and/or situations when eye protection, proper lab attire, etc. are not necessary.
	Know the rules for being trained on and authorized to use the lab's specialized equipment, e.g., centrifuge, rotary evaporator, glove box, etc.	In most cases this equipment should have a separate SOP which can be used in conjunction with any owner manuals as a training tool for that piece of equipment.
	Know the lab's "Do's and Don'ts"	For example, what are the lab rules regarding propping open lab doors, food storage, break areas, working after hours, cleaning up after yourself, etc. Refer to SU's Laboratory Toolkit: https://ehs.stanford.edu/topic/lab-safety
EMERGENCY EQUIPMENT & PROCEDURES	Know where to find safety equipment.	Includes spill kits, fire extinguishers, emergency alarm boxes, safety eyewash and showers, and first aid kits. List safety equipment relevant to your lab in this box and either describe location or as part of training show new lab personnel the location during lab walkthrough
	Know the emergency procedures for different hazards, including biohazards, chemical hazards, radioactive hazards, fire, and earthquakes.	Include: - What equipment do I need to quickly turn off before evacuating (heat sources, gases, vacuums, etc.)? - Where is the Emergency Assembly Point (EAP)? - What are at least two evacuation routes out of the building?
	Know the incident and injury reporting procedures.	Include: - How to obtain and complete an incident investigation report (SU-17 form) via su17.stanford.edu - How to call 911 from a campus phone and cell phone - How to contact the SU Occupational Health Center (650) 725-5308
Lab member: _____ Lab member's signature: _____ Trainer's name: _____ Training date: _____ Signature of PI: _____		

ADDITIONAL LABORATORY-SPECIFIC OR JOB-SPECIFIC TRAINING LOG

Researcher Name: _____ PI/Laboratory Name: _____

Use this form to document additional training on specific laboratory procedures, laboratory equipment, physical hazards, or electrical hazards. Initial laboratory safety training should be documented using the [Stanford University Laboratory-Specific Training Checklist](#). Laboratory specific training on procedures involving **radioactive substances** or **lasers** requires separate documentation paperwork. Please contact Health Physics at 650-723-0448 for additional information.

Description of Training: _____

☐ Classroom training ☐ Online training ☐ Instrument training ☐ Hands-on/procedure ☐ Other: _____

Trained by: _____ Trainee Signature: _____ Date: _____

Description of Training: _____

☐ Classroom training ☐ Online training ☐ Instrument training ☐ Hands-on/procedure ☐ Other: _____

Trained by: _____ Trainee Signature: _____ Date: _____

Description of Training: _____

☐ Classroom training ☐ Online training ☐ Instrument training ☐ Hands-on/procedure ☐ Other: _____

Trained by: _____ Trainee Signature: _____ Date: _____

Description of Training: _____

☐ Classroom training ☐ Online training ☐ Instrument training ☐ Hands-on/procedure ☐ Other: _____

Trained by: _____ Trainee Signature: _____ Date: _____

Description of Training: _____

☐ Classroom training ☐ Online training ☐ Instrument training ☐ Hands-on/procedure ☐ Other: _____

Trained by: _____ Trainee Signature: _____ Date: _____

Description of Training: _____

☐ Classroom training ☐ Online training ☐ Instrument training ☐ Hands-on/procedure ☐ Other: _____

Trained by: _____ Trainee Signature: _____ Date: _____

Description of Training: _____

☐ Classroom training ☐ Online training ☐ Instrument training ☐ Hands-on/procedure ☐ Other: _____

Trained by: _____ Trainee Signature: _____ Date: _____

Description of Training: _____

☐ Classroom training ☐ Online training ☐ Instrument training ☐ Hands-on/procedure ☐ Other: _____

Trained by: _____ Trainee Signature: _____ Date: _____

Description of Training: _____

☐ Classroom training ☐ Online training ☐ Instrument training ☐ Hands-on/procedure ☐ Other: _____

Trained by: _____ Trainee Signature: _____ Date: _____

Description of Training: _____

☐ Classroom training ☐ Online training ☐ Instrument training ☐ Hands-on/procedure ☐ Other: _____

Trained by: _____ Trainee Signature: _____ Date: _____

Laboratory Personal Protective Equipment (PPE) Assessment Tool

This tool is designed to:

- Identify and document hazards in your lab and the required PPE to minimize exposure.
- Document the required completion of laboratory-specific PPE training.

The Supervisor or PI/Lab Supervisor may assign a designee to perform or assist in the above duties but must ensure they are carried out.

Section 1: Laboratory Information

Department	
Principal Investigator/ Lab Supervisor	
Phone Number, e-mail	
Building/Rooms	

Section 2: PPE Assessment

1. Assess potential hazards and appropriate PPE for laboratory operations under your supervision.
 - a. Include a walk-through survey of lab areas.
 - b. Use the checklist below.
 - Check the corresponding boxes for the operations/tasks your lab conducts.
Note: If a risk assessment by the PI/Lab Supervisor determines that, for some operations, more or less protective PPE is required than the applicable PPE indicated, this must be documented on the PPE assessment tool, the appropriate Standard Operating Procedure (SOP), or a separate document.
 - Use the section at the end of the checklist to include any hazards and PPE not covered in the form.
2. Complete and sign the "Certification of PPE Assessment." Keep a local copy.
3. Update the PPE assessment when new hazards are introduced into your work area.

For further technical guidance and assistance with PPE selection, consult with EH&S at 723-0448.

Chemical Hazards		
Activities performed in the lab?	Lab Operation/Task Involving ¹	Applicable PPE (in addition to proper street clothing ²)
☐ Yes ☐ No	Flammable liquids	• Safety glasses (If splash potential exists, use goggles + face shield instead.) • Lab coat (A flame-resistant lab coat, such as Nomex, may be appropriate depending upon the quantity [> 4L] or the task, e.g., heating.) • Appropriate chemical-resistant gloves

Laboratory Personal Protective Equipment (PPE) Assessment Tool

☐ Yes ☐ No	Corrosive liquids	- Safety glasses (If splash potential exists, use goggles + face shield instead.) - Lab coat (Also use chemical-resistant apron if splash potential exists.) - Appropriate chemical-resistant gloves
☐ Yes ☐ No	Cryogenic liquids or dry ice (including working with cryogenic dewars)	- Safety glasses (If splash potential exists, use goggles + face shield instead.) - Lab coat - Insulated cryogenic gloves
☐ Yes ☐ No	Compressed Gases	- Safety glasses - Lab coat, as needed (e.g., when making or breaking connections with non-inert gases) - Gloves, as needed (e.g., work gloves when handling cylinders, chemical-resistant gloves when making or breaking connections with non-inert gases)
☐ Yes ☐ No	Pyrophoric or water reactive compounds (or highly exothermic reactions)	- Goggles + face shield - Flame-resistant lab coat, such as Nomex - Appropriate chemical-resistant gloves (Additional fire resistant gloves may be necessary, depending on the task.) - Non-synthetic street clothing
☐ Yes ☐ No	Explosive compounds	- Goggles + face shield - Flame-resistant lab coat, such as Nomex - Heavyweight gloves, such as anti-static PVC gauntlets - Engineering control: Use blast shield.
☐ Yes ☐ No	Engineered nanomaterials	- Safety glasses (If splash potential exists, use safety goggles + face shield instead.) - Disposable [®]Tyvek-type coveralls (or Lab coat) - Appropriate chemical-resistant gloves - For additional guidance, see Engineered Nanomaterials.
☐ Yes ☐ No	Particularly hazardous substances, including select carcinogens , reproductive toxins , and substances with a high degree of acute toxicity	- Safety glasses (If splash potential exists, use safety goggles + face shield instead.) - Lab coat - Appropriate chemical-resistant gloves
☐ Yes ☐ No	Chemically preserved animal and/or human specimens	- Safety glasses - Gown or lab coat - Appropriate chemical-resistant gloves
☐ Yes ☐ No	Hazardous chemical not in one of the above special categories	- Safety glasses (If splash potential exists, use goggles + face shield instead.) - Lab coat - Appropriate chemical-resistant gloves

Biological Materials

Activities performed in the lab?	Lab Operation/Task Involving ¹	Applicable PPE (in addition to proper street clothing ²)
☐ Yes ☐ No	Working with biological agents or recombinant DNA classified as Biosafety Level 1	No PPE required. However, if working in conjunction with another hazard (e.g., flammable liquids), wear appropriate PPE for that hazard.

Laboratory Personal Protective Equipment (PPE) Assessment Tool

☐ Yes ☐ No	Working with biological agents or recombinant DNA classified as Biosafety Level 2	• Safety glasses (If splash potential exists, use safety goggles and face shield.) <i>Note: When using the Biological Safety Cabinet (BSC), eye and face protection is not required for work with biological agents or recombinant DNA.</i> • Lab coat • Latex or nitrile gloves
☐ Yes ☐ No	Working with infectious agents or recombinant DNA classified as Biosafety Level 2+	• Safety goggles (If splash potential exists, also use face shield.) <i>Note: When using the Biological Safety Cabinet (BSC), eye and face protection is not required for work with biological agents or recombinant DNA.</i> • Disposable gown or lab coat • Latex or nitrile gloves • Respirator (as determined by Administrative Panel on Biosafety [APB] protocol review; contact EH&S for assessment 723-0448)
☐ Yes ☐ No	Working with Infectious agents or recombinant DNA classified as Biosafety Level 3	• Safety goggles (If splash potential exists, also use face shield.) <i>Note: When using the Biological Safety Cabinet (BSC), eye and face protection is not required for work with biological agents or recombinant DNA.</i> • Full disposable gown or [®]Tyvek suit • Shoe cover or dedicated shoes • Latex or nitrile gloves (double) • Respirator (as determined by the APB review; contact EH&S for assessment, 723-0448)
☐ Yes ☐ No	Human or non-human primate blood and other body fluids, tissues or cells, or blood borne pathogens (BBP)	• Safety glasses (If splash potential exists, use safety goggles and face shield.) <i>Note: When using the Biological Safety Cabinet (BSC), eye and face protection is not required for work with biological agents or recombinant DNA.</i> • Lab coat • Latex or nitrile gloves
☐ Yes ☐ No	Live Animals (Animal Biosafety Level 1 or Risk Category 2 animals)	• Safety glasses (If splash potential exists, use safety goggles + face shield.) <i>Note: When using the Biological Safety Cabinet (BSC), eye and face protection is not required for work with biological agents or recombinant DNA.</i> • Lab coat • Latex, nitrile, or vinyl gloves (+ wire mesh gloves as appropriate) • Consult with EH&S for N95 respirator assessment, 723-0448 • In the space provided at the end of the checklist, list any additional or modified PPE required by (1) an APB or APLAC protocol or (2) consultation or written policies from VSC and/or EH&S (723-0448).
☐ Yes ☐ No	Live Animals (Animal Biosafety Level 2 or Risk Category 1 animals)	• Safety glasses (If splash potential exists, use safety goggles + face shield.) <i>Note: When using the Biological Safety Cabinet (BSC), eye and face protection is not required for work with biological agents or recombinant DNA.</i> • Disposable gown, hair cover, shoe cover, and surgical mask • Latex, nitrile, or vinyl gloves (+ wire mesh gloves as appropriate) • In the space provided at the end of the checklist, list any additional or modified PPE required by (1) an APB or APLAC protocol or (2) consultation or written policies from VSC and/or EH&S (723-0448).

Laboratory Personal Protective Equipment (PPE) Assessment Tool

Radiation		
Activities performed in the lab?	Lab Operation/Task Involving ¹	Applicable PPE (in addition to proper street clothing ²)
☐ Yes ☐ No	Unsealed radioactive materials or waste	For radionuclide-specific PPE, consult with Health Physics, 723-3201.
☐ Yes ☐ No	Class 3B or 4 laser	Appropriate laser safety goggles (consult with Health Physics, 723-3201)
☐ Yes ☐ No	Laser(s) modified by optics	
☐ Yes ☐ No	Ultraviolet (UV) radiation (from sources other than lasers)	- UV-blocking eye protection or UV-blocking face shield - Appropriate UV blocking gloves
☐ Yes ☐ No	Infrared-emitting equipment	Appropriately-shaded goggles for infrared radiation

Physical Hazards		
Activities performed in the lab?	Lab Operation/Task Involving ¹	Applicable PPE (in addition to proper street clothing ²)
☐ Yes ☐ No	Glassware (or other vessels) under pressure or vacuum	- Safety goggles + face shield - Chemical-resistant apron for high risk activities - Appropriate chemical-resistant gloves - Engineering control: Use blast shield, as appropriate.
☐ Yes ☐ No	Working with knives, scalpels, razor blades, etc. or handling broken glass	Cut-resistant gloves
☐ Yes ☐ No	Centrifuge	- Safety glasses - If centrifuging hazardous materials, wear additional PPE to match the hazard.
☐ Yes ☐ No	Sonicator or other loud equipment	Ear plugs or ear muffs may be required (consult with EH&S 723-0448).
☐ Yes ☐ No	Removing freezer vials from liquid nitrogen	- Goggles + face shield - Lab coat - Insulated cryogenic gloves
☐ Yes ☐ No	Handling hot liquids/equipment (e.g., autoclaved materials, heated glassware, water or oil bath)	- Safety glasses (If splash potential exists, use safety goggles + face shield.) - Lab coat - Thermally insulated gloves (Wear chemical-resistant gloves underneath, as needed.)
☐ Yes ☐ No	Machinery (e.g., lathes, saws) and hand tools	- Safety glasses (+ face shield if flying fragments or particles generated) - Gloves appropriate for hazards (e.g., chemicals, sharp objects) Note: Gloves are not required if there is a potential to become entangled in moving parts; consult with EH&S 723-0448. - Work Practice: Confine long hair/beards to prevent entanglement in machinery (e.g., via bun, pinned-up ponytail, or hairnet). Do not wear any loose clothing and jewelry. - Hearing protectors, respiratory protection, or safety shoes may be required (consult with EH&S 723-0448). - See SU Safe Operation of Shop Machinery for additional PPE Guidance.

Laboratory Personal Protective Equipment (PPE) Assessment Tool

Other Laboratory Operations/Tasks	
Lab Operation/Task Involving:	Applicable PPE:
Personnel that are not directly involved in the lab operations, but who are at risk for potential exposure to hazardous materials and/or physical hazards	- Lab coat - Safety glasses - Additional PPE deemed necessary (e.g., if touching contaminated surfaces, wear gloves to match the hazard) - Proper street clothing - long pants (or equivalent) that cover legs and ankles, and non-perforated, closed-toed shoes that completely cover the feet

¹ Conduct activities with potential to generate airborne contaminants using appropriate engineering controls (e.g., laboratory fume hood, biosafety cabinet, glove box, local exhaust at work bench). If engineering controls are not feasible, consult EH&S to determine if the activity presents a respiratory hazard, which may require a respirator; call 723-0448.

² Proper street clothing - *Long pants (or equivalent) that cover the legs and ankles, and non-perforated, closed-toed shoes that completely cover feet.*

Certification of PPE Assessment	
Name of person conducting assessment	
Title	
Phone Number, email	
Signature:	Date:

Laboratory Personal Protective Equipment (PPE) Assessment Tool

Section 3: Laboratory-Specific PPE Training

1. Deliver laboratory-specific PPE training

Train lab personnel upon their joining the lab or prior to performing work requiring the use of PPE. Training content shall include, but not be limited to:

- When and what PPE is required per *Section 2: PPE Assessment*
- Limitations of the PPE
- How to properly put on, adjust, wear, and remove PPE
- Proper care, maintenance, useful lifespan, and disposal of PPE

For more information, refer to the PPE Safety Training Guidance at <http://ehs.stanford.edu/resource/ppe-training-guidance>

2. Training documentation

- When lab personnel have demonstrated an understanding of the above training and ability to the use PPE properly, the lab member and trainer must sign below that the PPE training has been conducted.
- Maintain training records for at least one year.

3. Provide retraining

Retraining is required of laboratory personnel when:

- Changes in laboratory activities/operations render previous PPE training obsolete.
- Inadequacy of laboratory personnel's knowledge or use of PPE is evident.

PPE Training Verification

Trainee Name:	Trainee Signature:	SUNET ID:	Training Date:	Trainer Name:	Trainer Signature:

STANDARD OPERATING PROCEDURES (SOPs) & RISK ASSESSMENT

Restricted Chemicals and High Risk Procedures

<https://ehs.stanford.edu/topic/chemical-safety/restricted-chemicals-high-risk-procedures>

Prior approval is the process whereby laboratory personnel seek permission, and the PI or lab supervisor grants approval, for the use of restricted chemicals. Restricted chemicals that require prior approval include:

- Toxic gases regulated by Santa Clara County (e.g. diazomethane, hydrogen cyanide, hydrogen fluoride (anhydrous), and nickel carbonyl)
- Dimethylmercury
- Chemicals classified as Department of Transportation (DOT) Division 1.1 explosives (e.g., picric acid, 2,4-dinitrophenylhydrazine (DNPH), ammonium perchlorate)
- Tert-butyllithium
- Hydrofluoric acid
- Tetramethylammonium Hydroxide
- Procedures with “high” and “unacceptable” risk ratings on a self-conducted [Laboratory Risk Assessment](#)

Lab personnel should consult with the PI or their supervisor on certain higher-risk chemical usage and operations so that special safety precautions can be taken where appropriate.

Consultation can include discussion of special hazards and safety precautions and review of applicable standard operating procedures.

What to do?	How to do it?
For the use of restricted chemicals, follow these steps to obtain prior approval	1. Complete the Documenting SOP Review and PI Approval, or have the PI or lab supervisor sign and date the staff member’s notebook and indicate approval for the process, procedure, or activity (or have the PI or supervisor use other written approval, such as an e-mail or memo). 2. Retain record of prior approval for at least one year.

SOP Guidance

<https://ehs.stanford.edu/sop-guidance>

Standard Operating Procedure

A standard operating procedure (SOP) is a set of written instructions that describes in detail how to perform a process or experiment safely and effectively. SOPs may be utilized in laboratory and non-laboratory operations. Within the laboratory environment, SOPs are a requirement of Stanford University's [Chemical Hygiene Plan](#), as described in section 4.

EH&S is available for consultation on SOPs. Submit your request for EH&S review of an SOP by completing the [SOP Intake Form](#). Please note that EH&S reviews an SOP but does not replace PI/lab supervisor review and approval.

Examples of SOP topics:

Lab Areas:

- The specific use of a chemical or class of chemicals (such as a specific laboratory procedure).
- The generic use specific chemical or class of chemicals with similar hazards (for example, mineral acids).
- A generic procedure (such as distillation) that covers several chemicals.
- Radionuclide use in DNA and RNA labeling.
- Laser operations.
- X-ray diffraction machine operation.
- Use and maintenance of laboratory equipment or instruments.

Non-lab Areas:

- Specific work practices (such as lifting techniques) to mitigate physical/health hazard.
- Specific tasks that are unique to work performed at Stanford University.

Responsibility in Laboratories

The PI/Lab Supervisor is responsible for providing written SOPs relevant to health and safety for laboratory activities he/she directs involving hazardous chemicals. Laboratory personnel working autonomously or performing independent research are responsible for developing SOPs appropriate for their own work using the guidance below.

Prioritizing SOP Development for Laboratories

Priority for SOP development should be given to any operation involving Restricted Chemicals, certain higher risk chemicals, such as Particularly Hazardous Substances and highly reactive chemicals, and specified higher risk research procedures described in the CHP.

Guidance for Non-Laboratory Areas

The Supervisor is responsible for providing written SOPs relevant to health and safety for activities he/she directs that require specific work practices to mitigate physical and/or health hazards.

Prioritizing SOP Development in non-lab areas

Priority for SOP development should be given to any operation involving control of energy sources via lockout/tagout, confined space entry, fall protection use, and/or respirator use.

What to do?	How to do it?
Prioritize SOPs to generate	Consult with PI/Laboratory Supervisor/Supervisor
Create SOPs	SU SOP Template SU SOP Template for Alkali Metals SU SOP Template for Hydrofluoric Acid / Fluoride solutions SU SOP Template for Hazardous Chemicals in Rodents SU SOP Template for Electrical, Engineering, and Robotics Research Work SU SOP Template for Nanomaterials SU SOP Template for Toxic Gases SU SOP Template for tert-Butyllithium SU Non-lab Task Safety Analysis Template
Incorporate General Use SOPs	Carcinogens Compressed Gases Corrosive Materials Cryogenic Liquids Flammable & Combustible Liquids Highly Acutely Toxic Materials Highly Reactive/Unstable Materials Irritants Reproductive Toxins Sensitizers
Maintain SOPs	Maintain copies of SOPs that are accessible to all lab members
Revise SOPs	Consult with PI/lab supervisor on a timeframe for reevaluating and revising SOPs

EH&S is available for consultation on SOPs. Submit your request for EH&S review of an SOP by completing the [SOP Intake Form](#). Please note that EH&S reviews an SOP but does not replace PI/lab supervisor review and approval.



STANDARD OPERATING PROCEDURE TEMPLATE

#1	CONTACT INFORMATION	
Procedure Title		[Specify – Note: All guidance text in brackets may be deleted]
Procedure Author		[Specify]
Creation/Revision Date		[Specify]
Responsible Person		[Name of PI, Lab Supervisor, or Autonomous Researcher, as appropriate]
Location of Procedure		[Building and room number]
Approval Signature		[Obtain prior approval, as appropriate. See section #10 of this template.]
#2	THIS STANDARD OPERATING PROCEDURE (SOP) IS FOR A:	
☐ Specific laboratory procedure or experiment [Examples: synthesis of chemiluminescent esters, folate functionalization of polymeric micelles, etc.]		
☐ Generic laboratory procedure that covers several chemicals [Examples: distillation, chromatography, etc.]		
☐ Generic use of specific chemical or class of chemicals with similar hazards [Examples: organic azides, mineral acids, etc.]		
#3	PROCESS OR EXPERIMENT DESCRIPTION	
[Provide a brief description of your process or experiment, including its purpose. Do <u>not</u> provide a detailed sequential description as this will be covered by section #6 of this template. Indicate the frequency and duration below.]		
Frequency:		☐ one time ☐ daily ☐ weekly ☐ monthly ☐ other: _____
Duration per Expt:		_____ minutes; or _____ hours
#4	SAFETY LITERATURE REVIEW & HAZARD SUMMARY	

1. Hazardous Substances

[List hazardous substances and their associated health and safety hazards. Examples of potential hazards include toxicity, reactivity, flammability, corrosivity, pressure, etc. Refer to Safety Data Sheets (SDSs) and other resources, as needed.]

2. Other Hazards

[List nonchemical hazards, e.g., biological hazards, electrical hazards, mechanical hazards, nonionizing radiation, or ionizing radiation.]

3. References

[List all references you are using for the safe and effective design of your process or experiment, including safety literature and peer-reviewed journal articles. Safety resources are available at <http://web.stanford.edu/dept/EHS/cgi-bin/lcst/creating-standard-operating-procedures/>.]

#5

STORAGE REQUIREMENTS

[Describe special handling and storage requirements for hazardous chemicals in your laboratory, especially for highly reactive/unstable materials, highly flammable materials, and corrosives.]

#6

STEP-BY-STEP OPERATING PROCEDURE

[For each step's description, include any step-specific hazard, personal protective equipment, engineering controls, and designated work areas in the left hand column.

- a. **Guidance on Engineering and Ventilation Controls** – Review safety literature and peer-reviewed journal articles to determine appropriate engineering and ventilation controls for your process or experiment. Guidance is available from health and safety specialists at Stanford EH&S and online in the General Use SOPs and Laboratory Safety Sheets in the Laboratory Chemical Safety Toolkit (<http://chemtoolkit.stanford.edu/>)
- b. **Guidance on Personal Protective Equipment** - To assist with your PPE selection, refer to <http://chemtoolkit.stanford.edu/LabPPE>. Respiratory protection is generally not required for lab research, provided the appropriate engineering controls are employed. For additional guidance on respiratory protection, consult with EH&S, 723-0448.
- c. **Designated work area(s)** - Required whenever carcinogens, highly acutely toxic materials, or reproductive toxins are used. The intent of a designated work area is to limit and minimize possible sources of exposure to these materials. The entire laboratory, a portion of the laboratory, or a laboratory fume hood or bench may be considered a designated area. See the Chemical Safety Toolkit for more information.

Describe the possible risks involved with failure to follow a step in the SOP in the right hand column.]

Step-by-Step Description of Your Process or Experiment	Potential Risks if Step is Not Done or Done Incorrectly (if any)
1. Don personal protective equipment. ☐ appropriate street clothing (long pants, closed-toed shoes)	

☐ gloves; indicate type: _____ ☐ safety goggles ☐ safety glasses ☐ face shield ☐ lab coat ☐ flame-resistant lab coat ☐ other: _____	
2. Check the location/accessibility/certification of the safety equipment that serves your lab:	
ITEM	STATUS
Laboratory Fume Hood/Glove Box or other Ventilation Control	Location: _____ <i>Check sticker to ensure that hood was certified within last 12 months.</i>
Eyewash/Safety Shower	Location: _____ <i>Ensure that it is accessible, not blocked. Check tag that it has been tested within last month.</i>
First Aid Kit	Location: _____
Chemical Spill Kit	Location: _____
Fire Extinguisher	Location: _____
Telephone	Location: _____
Fire Alarm Manual Pull Station	Location: _____
3. [Describe the next step in the procedure.]	
4. [Describe the next step in the procedure. Insert additional rows in table, as needed.]	
5. Dispose of hazardous solvents, solutions, mixtures, and reaction residues as hazardous waste.	
6. Clean up work area and lab equipment. [Describe specific cleanup procedures for work areas and lab equipment that must be performed after completion of your process or experiment. For carcinogens and reproductive toxins, designated areas must be immediately wiped down following each use.]	
7. Remove PPE and wash hands.	

#7	EMERGENCY PROCEDURES
1. Health-Threatening Emergencies A. Fire, explosion, health-threatening hazardous material spill or release, compressed gas leak, or valve failure, etc.	

- 1) Call 9-911 (or 286 in the School of Medicine).
- 2) Alert people in the vicinity and activate the local alarm systems.
- 3) Evacuate the area and go to your Emergency Assembly Point (EAP): [Indicate EAP location]
- 4) Remain nearby to advise emergency responders.
- 5) Once personal safety is established, call EH&S at 725-9999 (or 286 in the School of Medicine).
- 6) Provide local notifications (local notifications are listed at the end of this section).

Note: For compressed gas leaks, shut off gas supply only if this can be done safely, without risk to personnel.

B. Injuries and Exposures:

- 1) Remove the injured/exposed individual from the area, unless it is unsafe to do so because of the medical condition of the victim or the potential hazard to rescuers.
- 2) Call 9-911 (or 286 in the School of Medicine) if immediate medical attention is required.
- 3) Call 725-9999 (or 286 in the School of Medicine) to report the exposure to EH&S.
- 4) Administer first aid as appropriate.
- 5) Flush contamination from eyes/skin using the nearest emergency eyewash/shower for a minimum of 15 minutes. Remove any contaminated clothing.
- 6) Bring to the hospital copies of SDSs for all chemicals the victim was exposed to.

2. Non-Health-Threatening Emergencies

A. Injuries and Exposures

For injuries and exposures that are not considered [serious or a medical emergency](#), call the Occupational Health Center (OHC) at 725-5308 between 8:00 am-5:00 pm M-F for immediate phone triage and to schedule an appointment. For [urgent conditions](#) when SUOHC is closed, go to the Stanford University Medical Center Emergency Department.

B. Spills

For hazardous material spills or releases which have impacted the environment (via the storm drain, soil, or air outside the building) or for a spill or release that cannot be cleaned up by local personnel:

- 1) Notify Stanford University responders by calling 725-9999 (or 286 in the School of Medicine). These services are available 24 hours a day, 7 days a week.
- 2) Provide local notifications (local notifications are listed at the end of this section).

3. Local Cleanup of Small Spills

In the event of a minor spill or release that can be safely cleaned up by local personnel using readily available equipment (absorbent available from EH&S in Small Spill Kit) and laboratory PPE:

- 1) Notify personnel in the area and restrict access. Eliminate all sources of ignition.
- 2) Review the SDS for the spilled material, or use your knowledge of the hazards of the material to determine the appropriate level of protection (do not clean up spills requiring respiratory protection locally).
- 3) Wearing appropriate personal protective equipment, clean up spill. Collect spill cleanup materials in a tightly closed container. Manage spill cleanup debris as hazardous waste.

- 4) Submit online waste pickup request to EH&S.
- 5) Reporting Requirements: All spills cleaned up locally must be reported if they occur outside of secondary containment. A spill that occurs within secondary containment (a laboratory hood is considered secondary containment) must be reported if it is greater than 30 ml or if it takes longer than 15 minutes to clean up. To report a spill, call EH&S at 725-9999 (or in the School of Medicine, x286) as soon as possible.

4. Lab-Specific Procedures

[This section is for any emergency procedures different from standard responses, or for additional emergency information due to the nature of materials or task. Include information on gas leaks, chemical spills, and personal exposure/medical emergency as appropriate.]

5. Building Maintenance Emergencies

Call Facilities Operations at 723-2281 (or 721-2146 in the School of Medicine) for building maintenance emergencies (e.g., power outages, plumbing leaks).

6. Local Notifications

[Identify the area management staff that must be contacted and include their work and after-hours numbers. This must include the principal investigator and may include the lab safety coordinator, facilities manager, and/or business manager.]

#8

WASTE DISPOSAL

[Describe the quantities of waste you anticipate generating and appropriate waste disposal procedures. Include any special handling or storage requirements for your waste. Contact EH&S at 723-0448 for questions and additional guidance.]

#9

TRAINING REQUIREMENTS

General Training *(check all that apply):*

- ☐ General Safety, Injury Prevention (IIPP) & Emergency Preparedness (EHS-4200)
- ☐ Chemical Safety for Laboratories (EHS-1900)
- ☐ Compressed Gas Safety (EHS-2200)
- ☐ Biosafety (EHS-1500)
- ☐ Life Sciences Research Laboratory Safety Training (EHS-4875)

[In the School of Medicine, EHS-4875 is required for laboratory personnel in lieu of EHS-1900, 2200, and 1500.]

- ☐ Other: _____

[Depending on the hazardous materials and processes you will be working with in this SOP, additional safety training may be required by the University. To evaluate if additional safety training is required, go to

<http://web.stanford.edu/dept/EHS/prod/training/index.html>.]

Location Where Records Maintained:

Laboratory-specific training (*check all that apply*):

- ☐ Review of SDS for chemicals involved in process/experiment
- ☐ Review of this SOP
- ☐ Other: _____

**Location Where Records
Maintained:**

#10

PRIOR APPROVALS

[You **must** seek prior approval from your principal investigator (PI) or lab supervisor if you plan to use **restricted chemicals** (dimethylmercury and toxic gases regulated by Santa Clara County).

You should also consult your PI or lab supervisor if your experiments involve **high-risk chemicals and operations**, as special safety precautions may need to be taken. High-risk chemicals and operations may involve chemicals with a high level of acute toxicity, carcinogens, reproductive toxins, and highly reactive materials. For additional guidance, see section 5.3 of the Chemical Hygiene Plan.

Your PI or lab supervisor's prior approval may be documented by his/her signature in the Approval Signature section of this document. For granting prior approval to individuals other than the procedure author, use one of the methods described at <http://web.stanford.edu/dept/EHS/cgi-bin/lcst/restricted-chemicals-high-risk-procedures/>.]

Prior Approval (*check if applicable*):

- ☐ Prior approval from the PI or lab supervisor is required for this procedure

Safety Fact Sheets

<https://ehs.stanford.edu/safety-fact-sheets>

Certain materials and procedures that are common and necessary in laboratories across campus have significant hazards associated with them. EH&S has authored fact sheets to assist laboratories in addressing these hazards. These fact sheets cover essential information on the hazards, safety protocols, recommended personal protective equipment, and emergency response procedures. These fact sheets can serve as starting points and references for laboratory personnel.

Hazard type	BIOLOGICAL	
Hazard	Adeno-Associated Virus Fact Sheet Adenovirus Fact Sheet Epstein-Barr Virus Fact Sheet Herpesvirus Fact Sheet Lentivirus Fact Sheet Moloney Murine Leukemia Virus Fact Sheet Pox Viruses/Vaccinia Fact Sheet Rabies Virus Fact Sheet Sendai Virus Fact Sheet	
Hazard type	PHYSICAL	
Hazard	Asbestos Fact Sheet Autoclave Safety Seguridad del Autoclave Centrifuge Safety Electrophoresis Safety Cryogenic Vial Safety Laboratory Ergonomics Safe Operation of Shop Machinery Shop Machinery Guidelines for Supervisors Warm & Cold Rooms – Safe Work Practices	Lead Bricks Microtome Soldering Safety
Hazard type	CHEMICAL	
Hazard	Information on Alkali Metals Information on Azide Compounds Information on Cyanide Compounds Information on Hydrofluoric Acid Information on Peroxide-Forming Compounds Information on Picric Acid Information on Piranha Solutions Lead Fact Sheet Nitrous Oxide Used for Laboratory Fume Hood Testing Safe Use of Anesthetic Gases in Laboratory Research Working with Air-Sensitive or Highly Reactive Compounds	4L Chemical Bottles Dental Acrylics Ethanol Hydrogenation Pyrophoric Compounds Sodium Hypochlorite (Bleach) Tetramethylammonium Hydroxide (TMAH) Base Bath Information
Hazard type	RADIOLOGICAL	
Hazard	Ag-110m Radionuclide Fact Sheet Am-241 Radionuclide Fact Sheet As-73 Radionuclide Fact Sheet As-74 Radionuclide Fact Sheet	

Ba-133 Radionuclide Fact Sheet
Be-7 Radionuclide Fact Sheet
Br-77 Radionuclide Fact Sheet
C-14 Radionuclide Fact Sheet
Ca-45 Radionuclide Fact Sheet
Cd-109 Radionuclide Fact Sheet
Ce-141 Radionuclide Fact Sheet
Cl-36 Radionuclide Fact Sheet
Co-57 Radionuclide Fact Sheet
Co-58 Radionuclide Fact Sheet
Co-60 Radionuclide Fact Sheet
Cr-51 Radionuclide Fact Sheet
Cs-137 Radionuclide Fact Sheet
F-18 Radionuclide Fact Sheet
Fe-55 Radionuclide Fact Sheet
Fe-59 Radionuclide Fact Sheet
Ga-67 Radionuclide Fact Sheet
Ga-68 Radionuclide Fact Sheet
Gd-153 Radionuclide Fact Sheet
Ge-68 Radionuclide Fact Sheet
H-3 Radionuclide Fact Sheet
Hg-203 Radionuclide Fact Sheet
I-123 Radionuclide Fact Sheet
I-125 Inorganic Radionuclide Fact Sheet
I-125 Organic Radionuclide Fact Sheet
I-131 Radionuclide Fact Sheet
In-111 Radionuclide Fact Sheet
Mn-54 Radionuclide Fact Sheet
Mo-99 Radionuclide Fact Sheet
Na-22 Radionuclide Fact Sheet
Nb-95 Radionuclide Fact Sheet
Ni-63 Radionuclide Fact Sheet
P-32 Radionuclide Fact Sheet
P-33 Radionuclide Fact Sheet
Pb-210/Bi-210/Po210 Radionuclide Fact Sheet
Rb-86 Radionuclide Fact Sheet
Re-186 Radionuclide Fact Sheet
S-35 Radionuclide Fact Sheet
Sb-125 Radionuclide Fact Sheet
Sc-46 Radionuclide Fact Sheet
Se-75 Radionuclide Fact Sheet
Sr82/Rb82 Generator Radionuclide Fact Sheet
Sr-89 Radionuclide Fact Sheet
Tc-99 Radionuclide Fact Sheet
Tc-99m Radionuclide Fact Sheet
Tl-201 Radionuclide Fact Sheet
U-Nat Radionuclide Fact Sheet
V-48 Radionuclide Fact Sheet
Various Radionuclides
Xe-133 Radionuclide Fact Sheet
Y-88 Radionuclide Fact Sheet
Y-90 Radionuclide Fact Sheet
Zn-65 Radionuclide Fact Sheet

Risk Assessment

<https://ehs.stanford.edu/topic/lab-safety/risk-assessment>

Evaluation and assessment of risk is an integral step in designing an experimental protocol. Not only does a thorough risk assessment allow researchers to systematically identify and control hazards, but also improves science through better planning and understanding of the research question and variables.

The Stanford Laboratory Risk Assessment Tool provides a framework for risk assessment that maps onto the scientific method, melding with the process researchers already use to answer scientific questions.

Stanford Laboratory Risk Assessment Tool

Contact EH&S at 650-723-0448 with any questions or to request support in conducting a risk assessment.

What are the steps for risk assessment?



The Risk Assessment Tool uses a four-part framework that can be used for an experiment, analytical process, or series of tasks:

- **Explore** – Determine the scope of your work, beginning with your research objective. What scientific question(s) are you trying to answer or problem(s) are you trying to solve? Conduct a broad review of the literature. Consider querying other disciplines. Talk with others who have done similar work.
- **Plan** – Outline your procedure/tasks. This may include a deeper dive into specific topics in the literature. Determine hazards associated with each step/task and the control measures for reducing risk. EH&S can help with more detailed guidance on how to control certain hazards.

- **Challenge** – What assumptions did you use? Question the importance of each step. Seek advice from others to challenge your thinking. Ask yourself “what could go wrong?” Assign a risk rating (revising steps or controls particularly if risk category is “high” or “unacceptable”). EH&S can provide experimental design review to help with this process.
- **Assess** – Implement a model, prototype, or trial run. Can you perform a dry run to familiarize yourself with equipment and procedures? Can you test your experimental design at a smaller scale or with a less hazardous material? Determine if any design changes are needed. Run your experiment and monitor how your controls perform. Assess as you go and make changes as necessary.

Who should do Risk Assessments?

All researchers should conduct a risk assessment prior to conducting an experiment for the first time. When repeating an experiment/process, researchers may refer to a previous risk assessment and update the “Assess” section with the findings.

The risk assessment process involves rating the risk of the experiment, from low to unacceptable risk. Consult with your PI/supervisor and EH&S if your risk rating is “high” or “unacceptable” to redesign the experiment and/or implement additional controls to reduce risk.

What is the PI/Supervisor’s role?

PIs and Lab Supervisors should review risk assessments conducted by lab personnel/researchers. When personnel identify a “high” or “unacceptable” risk, PIs must work with them to redesign experiments or implement controls to reduce risk. If the risk rating remains “high” or “unacceptable” after redesign, consult EH&S to provide guidance to control hazards.

In addition to review of risk assessments, PIs are also encouraged to use the risk assessment framework during group meetings to discuss planned or completed experiments.

When is a Risk Assessment used versus a Standard Operating Procedure (SOP)?

Risk Assessments and **Standard Operating Procedures (SOPs)** are complementary tools. An SOP is typically written for repeated operations, equipment use, or required maintenance. SOPs may also be required for high-hazard chemicals and operations (e.g. work involving **Restricted Chemicals**, **Particularly Hazardous Substances** and **Highly Reactive Chemicals**, and other high risk research procedures).

A Risk Assessment is well-suited to novel operations in which the researcher is designing and planning an experiment. However, the Risk Assessment Tool can also be helpful in understanding risk when preparing for procedures that have an established SOP, varying parameters (e.g. scale, reagents, conditions), or in preparing to write new SOPs.

Laboratory Risk Assessment Tool

The Stanford Laboratory Risk Assessment Tool provides a framework for risk assessment that maps onto the scientific method, melding with the process researchers already use to answer scientific questions.

This tool allows researchers to systematically identify and control hazards to reduce risk of injuries and incidents. Conduct a risk assessment prior to conducting an experiment for the first time.

The risk assessment process involves rating the risk of the experiment, from “low” to “unacceptable” risk. Consult with your PI/supervisor and EH&S if your risk rating is “high” or “unacceptable” to redesign the experiment and/or implement additional controls to reduce risk.



Procedure:			
Lab Group:			
Completed By:		Date:	



EXPLORE

Identify your research question and approach. What question are you trying to answer? What are you trying to measure or learn? What is your hypothesis? What approach or method will you use to answer your question? Are there alternative approaches?

Research Question(s)
Approach(s) or Method

Identify the general hazards (check all that apply). Perform background research to identify known risks of the reagents, reactions, or processes. Review protocols, [Safety Data Sheets](#) (SDSs), and safety information for hazardous chemicals, agents, or processes. Review accident histories within your laboratory/department and [Lessons Learned](#) at Stanford or other institutions.

Hazardous Agents

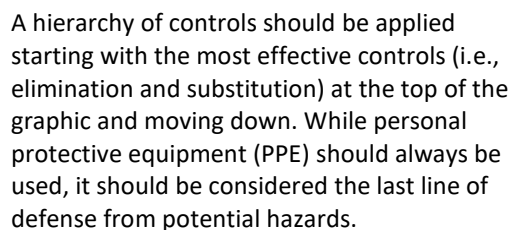
Physical Hazards of Chemicals	Health Hazards of Chemicals	Ionizing Radiation	Biohazards
☐ Compressed gases ☐ Cryogenics ☐ Explosives ☐ Flammables ☐ Organic peroxides ☐ Oxidizers ☐ Peroxide formers ☐ Pyrophorics ☐ Self-heating substances ☐ Self-reactive substances ☐ Substances which, in contact with water, emit flammable gases	☐ Acute toxicity ☐ Carcinogens ☐ Eye damage/irritation ☐ Germ cell mutagens ☐ Nanomaterials ☐ Reproductive toxins ☐ Respiratory or skin sensitization ☐ Simple asphyxiant ☐ Skin corrosion/irritation ☐ Specific target organ toxicity ☐ Hazards not otherwise classified	☐ Irradiator ☐ Radionuclide ☐ Radionuclide sealed source ☐ X-ray machine Non-Ionizing Radiation ☐ Lasers, Class 3 or 4 ☐ Lasers, Class 2 ☐ Magnetic fields (e.g., NMR, MRI) ☐ RF/microwaves ☐ UV lamps	☐ BSL-2 Biological agents ☐ BSL-3 Biological agents ☐ Human cells, blood, BBP ☐ NHPs/cells/blood ☐ Non-exempt rDNA ☐ Animal work ☐ High risk animals (RC1) ☐ Other (list):

Hazardous Conditions or Processes

Reaction Hazards	Hazardous Processes	Other Hazards
☐ Explosive ☐ Exothermic, with potential for fire, excessive heat, or runaway reaction ☐ Endothermic, with potential for freezing solvents decreased solubility or heterogeneous mixtures ☐ Gases produced ☐ Hazardous reaction intermediates/products ☐ Hazardous side reactions	☐ Generation of air contaminants (gases, aerosols, or particulates) ☐ Heating chemicals ☐ Large mass or volume ☐ Pressure > atmospheric ☐ Pressure < atmospheric ☐ Scale-up of reaction	☐ Hand/power tools ☐ Moving equipment/parts ☐ Electrical ☐ Noise > 80 dBA ☐ Heat/hot surfaces ☐ Ergonomic hazards ☐ Needles/sharps ☐ Other (list):

Field Hazards

Environmental Hazards	Site Hazards	Task/Equipment Hazards
☐ Foul weather ☐ Temperature extremes ☐ Intense sunlight ☐ Darkness/low light ☐ Altitude ☐ Smoke/dust ☐ Fire ☐ Animals/insects ☐ Plants/allergens ☐ Hygiene/water-borne and food-borne illness ☐ Vector-borne or other endemic diseases (list):	☐ Uneven/slippery surfaces ☐ Heights/drop-offs ☐ Falling objects ☐ Tight spaces/overhangs ☐ Boating/swimming/water hazards (waves, tides, current, depth) ☐ Navigation challenges ☐ Limited communication ☐ Remote area/limited medical services ☐ Personal security issues, risk of harassment or violence, US State Department active travel alert	☐ Driving/vehicle operation/traffic ☐ Lifting/carrying ☐ Digging/trenching ☐ Hand tools/power tools ☐ Sharp objects ☐ Strenuous physical activity ☐ Mental demands (e.g. long days, high stress environment, language barriers) ☐ Other (list):

[illegible]

² For guidance on selection of chemical-resistant gloves, see [Stanford's Laboratory Chemical Glove Selection Guide](#).

Select the appropriate PPE and safety supplies for the procedure (check all that apply).

Laboratory PPE/Safety Supplies

- | | |
|---|--|
| ☒ Appropriate street clothing (long pants, closed-toed shoes) | ☐ Fire extinguisher |
| ☐ Gloves; indicate type: _____ | ☐ Eyewash/safety shower |
| ☐ Safety glasses | ☐ First aid kit |
| ☐ Safety goggles | ☐ Spill kit |
| ☐ Face shield and goggles | ☐ Specialized medical supplies (e.g. calcium gluconate for hydrofluoric acid and amyl nitrite for cyanides) |
| ☐ Lab coat | |
| ☐ Flame-resistant lab coat | |
| ☐ Other (list): | |

Field PPE/Safety Supplies

- | | |
|--|---|
| ☒ Proper clothing (long pants, long sleeve shirt, warm layers, rain/wind protection, sun protection, hat etc.) | ☐ Sunscreen |
| ☒ Proper footwear (list): _____ | ☐ Anti-animal devices (e.g. bear bell, whistle, bear canister) |
| ☒ Communication device | ☐ Personal floatation device |
| ☐ Eye protection (safety glasses and/or sunglasses) | ☐ Fall protection |
| ☐ Work gloves | ☐ Road flares |
| ☐ Hardhat | ☐ Safety vests |
| ☐ Hearing protection | ☐ Extra food, water/water treatment method |
| ☐ First aid kit | ☐ Personal medications |
| ☐ Map (and GPS) | |
| | ☐ Other (list all): |

Identify the appropriate training (check all that apply). Identify the general safety and procedure based/specific training appropriate for your procedure.

General Safety Training

General/Chemical Safety

- ☒ General Safety & Emergency Preparedness (EHS-4200)
- ☐ Chemical Safety for Laboratories (EHS-1900)
- ☐ Compressed Gas Safety (EHS-2200)
- ☐ Cryogenic Liquids and Dry Ice Safety (EHS-2480)

Biosafety

- ☐ Biosafety (EHS-1500)
- ☐ Bloodborne Pathogens (EHS-1600) and Refresher (EHS-1601)

Radiation Safety

- ☐ Radiation Safety and Radiation Safety Hands-On (EHS-5250 and 5251)
- ☐ Radiation Safety SAIF (Small Animal Imaging Facility) (EHS-5255)
- ☐ Sealed Sources, Non-Irradiator (EHS-5265)
- ☐ Research Cabinet X-ray or Irradiator (EHS-1755) and Refresher (EHS-1756)
- ☐ Irradiator Security Training (EHS-4780) and Refresher (EHS-4781)
- ☐ Laser Safety (EHS-4820) and Refresher (EHS-4821)

Field Safety

- ☐ CPR
- ☐ Wilderness First Aid
- ☐ SCUBA certification/diving safety
- ☐ Driving safety
- ☐ Other (list):

Job Specific Training

- | | | |
|---|--|--|
| ☒ Lab/job-specific training | ☐ Emergency plans or field evacuation plans | ☐ Other (list): |
| ☐ Lab SOP(s) to review (list): | ☐ Equipment SOP(s) to review (list): | |



CHALLENGE

Question your methods. What have you missed and who can advise you? Challenge your hazard control measures by asking “What if...?” questions. “What if” questions should challenge you to find the gaps in your knowledge or logic. Factors to consider are human error, equipment failures, and deviations from the planned/expected parameters (e.g., temperature, pressure, time, flow rate, and scale/concentration).

What If Analysis

What if...? Examples: there is a loss of cooling? ...valves/stopcocks are left open/closed? ...there is unexpected over-pressurization? ...a spill occurs? ...the laser is misaligned? ...weather conditions change?

Then... ...there may be a runaway reaction. ...there may be an unexpected splash potential. ...the reaction vessel may fail. ...there may be a dermal exposure. ...there may be an eye injury. ...routes may be inaccessible.

What if...?

Then...

What if...?

Then...

What if...?

Then...

Assign a risk rating to the experiment. Based on your procedure outline and the what if analysis, determine the risk rating for the experiment or procedure.

Risk Rating: _____

Risk Rating Table¹

		Severity of Consequences - Personnel Safety			
		No Injuries	Minor	Moderate to life impacting	Life threatening from single exposure
Likelihood of Occurrence	(Almost) Certain	Low	High*	Unacceptable*	Unacceptable*
	Likely	Low	Medium	High*	Unacceptable*
	Possible	Low	Medium	High*	High*
	Rare	Low	Low	Medium	High*

¹The Risk Rating is subjective. The primary goal is for researchers to pause, think about risk, and differentiate unacceptable and high-level risk steps from those with a lower level risk. This will help drive additional consultation and control measures where needed.

Revise plan if the risk rating is too high.

Are these risks acceptable? Use the table below to determine the action to take based on the risk rating. What are the highest risk steps? What more can you do to control the risks? Return to planning and use the hierarchy of controls to design a safer experiment.

Risk Rating Action Table

Hazard Risk Rating	Action
Unacceptable*	STOP! Additional controls needed to reduce risk. Consult with PI and EH&S (650-723-0448).
High*	Additional controls recommended to reduce risk. Consult with PI and EH&S (650-723-0448).
Medium	Ensure you are following best practices. Consult with peers, PI, or EH&S, as needed.
Low	Perform work within controls.

PI/Supervisor Approval: _____

*Signature required for **High** risk ratings.

NOTE: **Unacceptable** risk rating experiments **may not proceed**. Introduce further controls to reduce risk.



ASSESS

Perform a trial run. How can you test your experimental design? Can you do a dry run of the procedure without hazardous chemicals/reagents/gases to familiarize yourself with equipment and demonstrate your ability to manipulate the experimental apparatus? Can you run the procedure with a less hazardous material? Can you test your experimental design at a smaller scale? If your procedure requires multiple people, would a table top exercise be useful?

Trial Run
Trial Run Procedure:
Did the trial go as expected? Yes ☐ No ☐
Experimental design changes needed (if any):

Perform and evaluate. Run your procedure using the appropriate controls you've identified. Evaluate controls and hazards as you work. Critique the controls and process you used by answering the following questions. If changes to controls are needed, update your risk assessment tool and re-evaluate any time you revise your process (e.g. changes in scale, reagent, equipment, or conditions that might increase the hazard/risk). Share your assessment with your PI/colleagues for the next iteration of the experiment.

Evaluate Your Procedure
What went well?
Did the controls perform as expected?
Did anything unexpected occur?
Did a hazard manifest itself that was not previously identified?
Were there any close-calls or near misses that indicate areas of needed improvement?
Did something go exceptionally well that others could learn from?
I plan to evolve my procedure by...

LABORATORY SELF-INSPECTIONS

Laboratory Inspections

<https://ehs.stanford.edu/topic/lab-safety/lab-inspections>

Laboratories must be inspected to identify and address potential safety and health deficiencies and to comply with regulations. There are many different types of inspections that must be completed. Use the table below to determine what type of inspection is required and how often it must take place.

What to do and where?	How to do it?
General Laboratory Self-Inspection For all laboratories	At least quarterly (more frequently where determined appropriate by PI or laboratory supervisor). • Use BioRaft (https://stanford.bioraft.com/) for lab self-inspections. • Retain records of inspection and any follow-up for at least three years.
Hazardous Materials Storage Area Self-Inspection For rooms designated as hazardous materials storage areas (including shared/common work areas and designated storage rooms)	At least monthly • Use BioRaft (https://stanford.bioraft.com/) for hazmat storage self-inspections or post the filled out paper form in the storage area. • Retain records of inspection and any follow-up for at least three years.
Controlled Substance Laboratory Self-Inspection For laboratories where controlled substances are used or stored (applicable to labs enrolled under the institutional program).	At least quarterly • Retain records of inspection and any follow-up for at least one year.
Laser Safety Self-Inspection For labs with lasers and laser use area.	At least annually • Retain records of inspection and any follow-up for at least one year • Submit self-inspection form to EHS Laser Safety Officer.
Shop Area Inspection For all locations, including labs, with shop machinery and tools	At least quarterly (more frequently where determined appropriate by PI/Laboratory Supervisor). • Use BioRaft (https://stanford.bioraft.com/) for lab inspections. • Retain records of inspection and any follow-up for at least 3 years.

[Testing frequencies of safety-related equipment \(eyewashes, fume hoods, etc.\)](#)

LABORATORY INSPECTION CHECKLIST

Building & Room:	Inspected By:
PI/Area Supervisor:	Date:

All laboratory spaces containing hazardous materials must be inspected at least quarterly. For each item check Yes, No, or N/A. Be sure to retain all documentation regarding inspections, including findings **and** corrective actions taken for any "No" responses, for a minimum of 3 years. Contact EH&S at 723-0448 for questions or additional information.

Y	N	N/A	GENERAL SAFETY
			1. Aisles and hallways are clear of obstructions for quick egress and equipment taller than 4 feet have earthquake bracing? (https://ehs.stanford.edu/manual/seismic-mitigation-guidance)
			2. Areas around fire extinguishers pull alarms and emergency eyewashes/showers clear and accessible?
			3. All objects stored at least 18 inches away from fire sprinklers?
			4. Extension cords and power strips not daisy-chained together and cords are not damaged? (https://ehs.stanford.edu/topic/general-workplace-safety/electrical-safety)
			5. Chemical workspaces organized and clean and free of food, drinks and cosmetics? (https://ehs.stanford.edu/topic/lab-safety)
Y	N	N/A	HAZARDOUS MATERIALS & WASTES
			6. Are all chemical hazardous waste containers labelled with a completed WasteTag (no containers should be labelled with only the word "Waste"? Other waste streams in the lab that are not hazardous or contaminated should be labeled - TRASH. (https://ehs.stanford.edu/topic/waste-disposal/handling-storing-waste)
			7. Can you visually verify that all hazardous waste containers have not been in the lab longer than 9 months? (https://ehs.stanford.edu/topic/waste-disposal/handling-storing-waste)
			8. Are all hazardous waste and hazardous material containers clean, structurally sound, closed when not in use, and stored in secondary containment? (https://ehs.stanford.edu/topic/waste-disposal/handling-storing-waste)
			9. Are all hazardous waste containers properly segregated according to hazard class? (https://ehs.stanford.edu/wp-content/uploads/SU_hazardouswasteposter.pdf)
			10. Is the "Chemical Waste Compliance" poster posted in the location where waste is stored? (https://ehs.stanford.edu/wp-content/uploads/SU_hazardouswasteposter.pdf)
			11. Chemical containers, including for non-hazardous chemicals and water, legibly labeled with the full chemical or trade name? (Note: abbreviations/formulas are not adequate.) (https://ehs.stanford.edu/topic/chemical-safety/chemical-inventory-management)
			12. Are all hazardous materials and oil pumps stored in secondary containment? (https://ehs.stanford.edu/topic/hazardous-materials/hazardous-materials-storage-requirements)
			13. Secondary containment clean and free from spilled material? (https://ehs.stanford.edu/topic/hazardous-materials/hazardous-materials-storage-requirements)
			14. Chemical containers properly segregated according to Stanford Segregation Scheme? (https://ehs.stanford.edu/forms-tools/stanford-storage-groups)
			15. Chemical inventory up-to-date in chemtracker including compressed gas cylinders? (https://ehs.stanford.edu/topic/hazardous-materials/hazardous-materials-storage-requirements)
			16. Flammable liquids (including flammable waste and glacial acetic acid) stored in flammable storage cabinets? (Note: Up to 10 gallons per control area (NOT

			individual lab) may be stored outside of cabinets.)
			17. All hazardous materials containers are closed and capped to minimize volatilization?
			18. Sharps placed in a designated sharps container and the container is no more than three-quarters full? (https://ehs.stanford.edu/wp-content/uploads/Lab-Medical-Waste-Guidance.pdf)
			19. Biohazardous waste in red bags in hard-sided container which is labeled with Universal Biohazard label on the top and lateral sides (must be labeled on all 4 sides and top)? (https://ehs.stanford.edu/wp-content/uploads/Lab-Medical-Waste-Guidance.pdf)
Y	N	N/A	COMPRESSED GASES
			20. Cylinders over 26 inches tall secured to a rigid structure at 1/3 and 2/3 height with metal chains, and a maximum of 2 cylinders used per pair of chains (one restraint for cylinders ≤26 inches and dewars)? (https://ehs.stanford.edu/manual/laboratory-standard-design-guidelines/storage-compressed-gas-cylinders-%E2%80%93-general)

Comments & Additional Findings (on following page)

Comments & Additional Findings

HAZARDOUS MATERIALS STORAGE AREA: MONTHLY INSPECTION

Building Number _____ Building Name _____ Room Number _____
 Inspector's Name _____ Phone Number _____ Year _____

Instructions:

1. Use this form to inspect rooms that are designated as Hazardous Materials Storage Areas (not laboratories or work areas).
2. Evaluate the storage area during the inspection for breakage, odors, etc. and for items listed below to ensure safe storage conditions.
3. Record all corrective actions in the last space for items with a "No" answer; attach an additional sheet if necessary.
4. If the spill is more than 30 ml or if it escapes the secondary container, call EH&S at **725-9999** (24 hours).

	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
Enter Inspection Date _____→												
Is the area free from leaks and/or spills? (Yes/No)												
Is appropriate emergency equipment (spill kit, etc.) available? (Y/N)												
Are incompatible chemicals segregated? (Y/N)												
Are all containers labeled with full chemical name? (Y/N)												
Are all containers closed? (Y/N)												
Are leak-proof secondary containers provided? (Y/N)												
Are gas cylinders and lecture bottles in storage properly restrained and valve caps in place? (Y/N/NA)												
After each inspection initial here _____→												
Corrective Action:												

BioRAFT User Instructions

Version Date: April 29, 2016

Updated: October 14, 2021

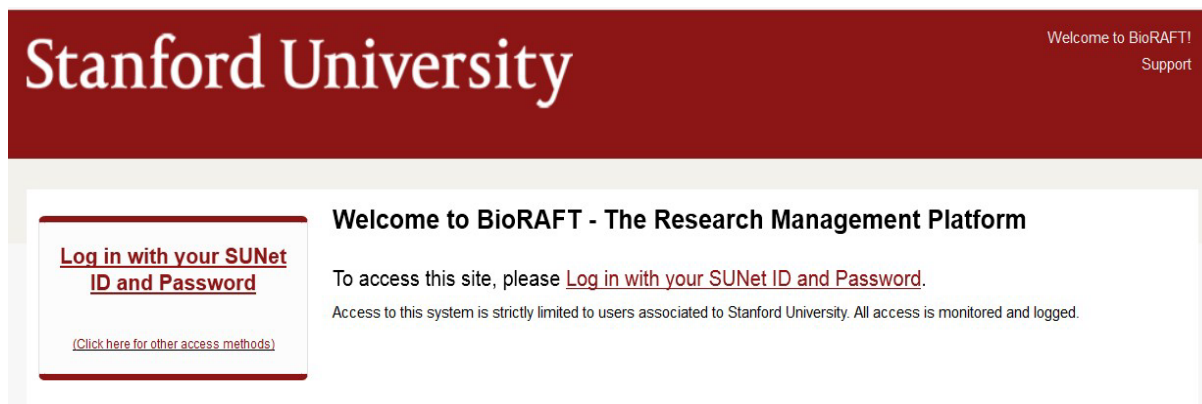
I. Introduction

BioRAFT is an electronic management tool provided by Stanford University to support a lab's health and safety efforts.

This document is a quick guide to familiarize users with basic BioRAFT functionalities. For further assistance with BioRAFT, please contact Environmental Health and Safety (EH&S) representatives at (650) 723-0448 or <https://ehs.stanford.edu/contact>.

II. BioRAFT Log in

The BioRAFT URL is <https://stanford.bioraft.com>.



Click on “[Log in with your SUNet ID and Password](#)” to proceed. Enter your SUNet ID and password, followed by two-step authentication.

III. Home Page

There are three sections to the research group's **Home page** in BioRAFT. Depending on the device being used (computer or mobile device) and the user's **permission** in BioRAFT, the BioRAFT **User Interface** may appear differently than what is displayed on the screen captures in this document.^{1, 2}

¹ The screen capture of the BioRAFT Homepage on the following page is as viewed on an iPad. All subsequent screen captures in this document show the BioRAFT User interface as viewed on a desktop or laptop computer.

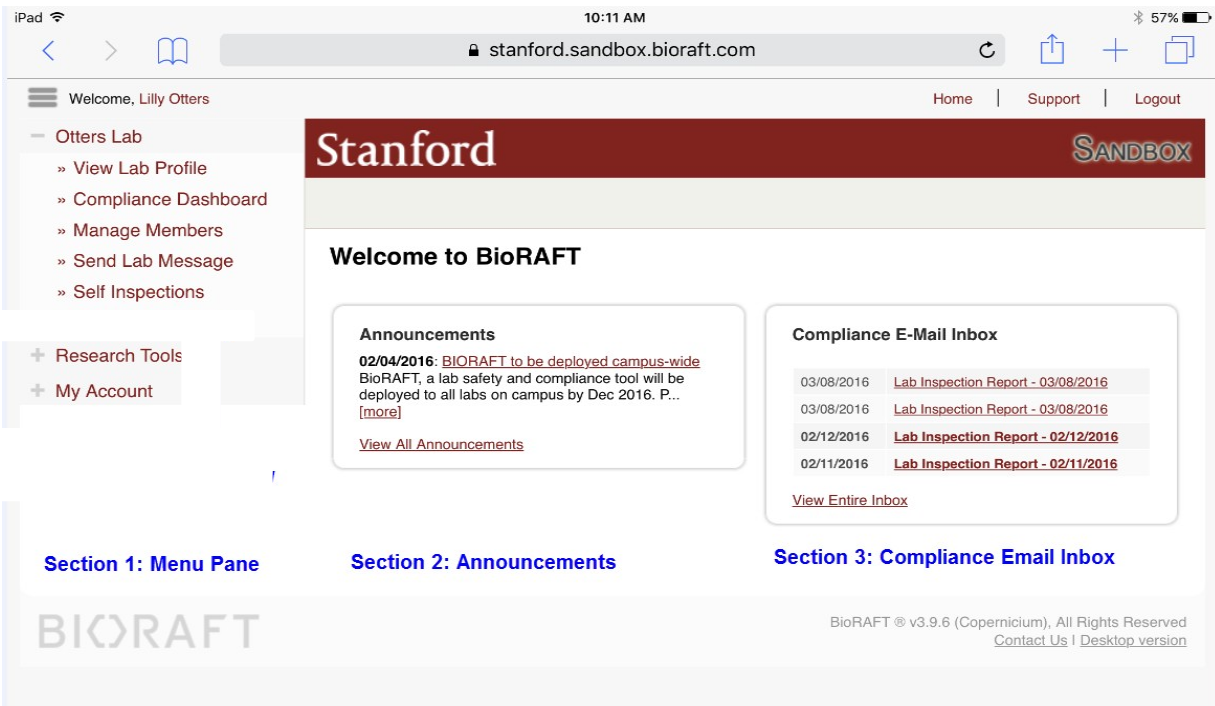
² Available menu items on the Menu Pane depends on user permission. For example, a user may see “View Members” instead of “Manage Members”.

Section 1: On the left side panel of the *Home* page is the **Menu Pane**. On the **Menu Pane**, users should see all the labs that they are a member of. In this document, screen captures are taken from “**Otters Lab**”, a fictitious lab for a fictitious Principle Investigator named Dr. Lilly Otters.

On an iPad, click on the  icon to expand or hide the **Menu Pane**. On a computer, the **Menu Pane** should automatically appear. Click on the “+” or “-” sign to expand or contract the menus on the pane.

Section 2: In the middle section are the **Announcements** posted by the EH&S department.

Section 3: On the right-hand side is the **Compliance E-mail Inbox**. It contains all correspondence sent via BioRAFT to the user’s email inbox. Click on an item in this section to view the email.



IV. Navigating in BioRAFT

To navigate between pages in BioRAFT, click on the available menu items on the page, or click on the **Back** or **Forward** button  in the browser.

V. Viewing Lab Information

1. To view lab information, click on **View Lab Profile** on the *Menu Pane* followed by the **View** tab on the horizontal menu bar as displayed on the screen shot below.
2. Below the **View** tab are menu items that users can view:
 - Summary
 - Spaces³
 - Forms (**This feature is currently disabled**)
 - Lab Inspections (for use by EH&S)
 - Self Inspections
3. To view an item, click on the item. For example, to view lab spaces, click on the **Spaces** item.

The screenshot shows the Stanford University Sandbox interface. At the top, the Stanford University logo is on the left, and the user's name 'Welcome, Lilly Otters' with links for 'Home', 'Support', and 'Logout' is on the right. Below the header, there is a horizontal menu bar with tabs: 'View', 'Edit', 'Dashboard', and 'Members'. The 'View' tab is selected. Below this, there is a sub-menu bar with items: 'Summary', 'Spaces', 'Documents', 'Forms', 'Lab Inspections', and 'Self Inspections'. The 'Spaces' item is highlighted. To the left of the main content area, there is a 'Menu Pane' for 'Otters Lab' with several options: 'View Lab Profile', 'Compliance Dashboard', 'Manage Members', 'Send Lab Message', 'Self Inspections', 'Manage Lab Forms', 'Research Tools', and 'My Account'. The 'View Lab Profile' option is highlighted. The main content area displays 'Otters Lab Spaces' with a table showing 2 results. The table has columns: 'Space Name', 'Description', 'Contact', 'Phone', and 'Groups/Labs'. The two rows represent 'ENVIRONMENTAL SAFETY FACILITY - 1100A' and 'ENVIRONMENTAL SAFETY FACILITY - 1110B', both with 'Jenny Y.O. Lam' as the contact and 'Otters Lab' as the group. Blue arrows and numbered circles (1, 2, 3) indicate the steps to view the lab spaces: (1) clicking 'View Lab Profile' in the menu pane, (2) clicking the 'View' tab, and (3) clicking the 'Spaces' item in the sub-menu bar.

Space Name	Description	Contact	Phone	Groups/Labs
ENVIRONMENTAL SAFETY FACILITY - 1100A	Example: Cell Culture Room	Jenny Y.O. Lam	650-222-2222	Otters Lab
ENVIRONMENTAL SAFETY FACILITY - 1110B	Example: Cell Culture Room (BSL2)	Jenny Y.O. Lam	650-222-2222	Otters Lab

³ Lab spaces cannot be entered or edited by the labs. Please contact EH&S to add, edit, or remove lab spaces.

VI. Editing Lab Information

1. To edit lab information, click on **View Lab profile** on the **Menu Pane** followed by the **Edit** tab on the horizontal menu bar.
2. Below the **Edit** tab are items that can be edited. For example, to edit the building name or room number, click on **Contact Info**.
3. To save the edits, click on the **Submit** button.

The screenshot displays the Stanford University Sandbox interface. At the top, the Stanford University logo is on the left, and a welcome message 'Welcome, Lilly Otters' with links for 'Home', 'Support', and 'Logout' is on the right. Below the header, a horizontal menu bar contains 'View', 'Edit', 'Dashboard', and 'Members'. The 'Edit' tab is active. On the left, a 'Menu Pane' for 'Otters Lab' lists options: 'View Lab Profile', 'Compliance Dashboard', 'Manage Members', 'Send Lab Message', 'Self Inspections', 'Manage Lab Forms', 'Research Tools', and 'My Account'. The 'View Lab Profile' option is highlighted with a blue box and a blue arrow labeled '1'. The main content area shows the 'Otters Lab' profile with tabs: 'Contact Info', 'Focus', 'Categories', 'Hazards', and 'Configure Defaults'. The 'Contact Info' tab is selected, indicated by a blue arrow labeled '2'. The form fields include: 'Building: *' (a dropdown menu showing 'ENVIRONMENTAL SAFETY FACILITY'), 'Room Number: *' (a text field with '8888' and a note to enter the primary room number), 'Mail Code:' (a text field with '1234'), 'Phone 1: *' (a text field with '650-111-1111' and a note to enter the primary contact number), 'Phone 2:' (a text field with '650-111-1111' and a note to enter the secondary contact number), 'Fax:' (a text field with '650-111-1111'), and 'description:' (a text area with 'Testing'). At the bottom, a 'Submit' button is highlighted with a blue arrow labeled '3'.

VII. Adding, Editing or Deleting Lab Members

To manage lab-member information, click on **Manage Members** on the **Menu Pane**. Please note that only members with the “Group Compliance Liaison” **Optional Access** will see this menu item. Others will see the **View Members** menu item instead. ⁴

1. To edit or remove member: Click on the **Edit** or **Remove** link next to the member’s name.
2. To add member: On the **Name** field, enter the member’s first name and last name, or SUNet ID. BioRAFT will search the Stanford directory for the member to be added.
3. Member’s Designation or Role: Click on the **Designation** field to bring up a designation/role list and select one.
4. Optional Access: Optional Access options under the **Optional Access in BioRAFT** section grants a member authority to perform certain tasks. Hover over the tool-tips ⓘ for information about each access and select one.
5. Job Activities: Select one or more **Job Activities** for the member. Hover over the tool-tips ⓘ for information about each activity.
6. When finished, click on the **Lookup/Add** button to complete adding the member.

Stanford University

Welcome, Lilly Otters
Home | Support | Logout

SANDBOX

View Edit Dashboard **Members**

Members | Manage Job Activities | Members History

Otters Lab Members

Last Confirmed: 03/21/2016 ⓘ

Name	Email	Designation	
Otters, Lilly	lillyotters@bioraft.com	Principal Investigator	Edit
Hanston, Ivy	ivyhanston@bioraft.com	Lab Manager	Edit Remove

[Confirm List](#)

Add a member

Name: (temporary account) Mark Tation ⓘ ⓘ

Designation: Lab Manager ⓘ

Optional Access in BioRAFT

- ☐ Edit Basic Group Information ⓘ
- ☐ Edit Group Equipment Inventory ⓘ
- ☒ Group Compliance Liaison ⓘ

Job Activities

General

- ☒ Works in Wet Laboratory spaces ⓘ
- ☐ Works in Dry Laboratory spaces ⓘ
- ☐ Works in Office Environments

[Lookup / Add](#) [Configure Defaults](#)

⁴ Permission to perform certain tasks in BioRAFT is tied to the **Optional Access** granted to the member.

VIII. SELF INSPECTION

Quarterly and monthly self inspections are an essential function to identify and address potential health and safety deficiencies and to fulfill regulatory compliance requirements. Lab spaces must be inspected quarterly whereas hazardous materials storage areas must be inspected monthly by each research group sharing the space.

All lab members can perform self inspections for their laboratory. The following sections describe how to view previous self inspections as well as how to initiate and perform a new self inspection.

A. Self Inspection Log

1. To view all recent self inspections performed, visit the laboratory **Self Inspection Log** by clicking on **Self Inspections** on the **Menu Pane**.
2. To search for a particular self inspection, enter search criteria on the search area as indicated on the screen shot below.
3. Findings from the self inspections conducted in the past 18 months will be displayed at the bottom of the page.
4. To conduct a self inspection, click on the **Add a new Self Inspection** link.

Stanford University Welcome, Lilly Otters
Home | Support | Logout
SANDBOX

View Edit Dashboard Members

Summary | Spaces | Forms | Lab Inspections | **Self Inspections**

Otters Lab Self Inspection Log

1 **Self Inspections**

2 **Search area**

Inspection Type: <All> Performed Between: Submit

[Add a new Self Inspection](#)

Date	Inspected Groups	Inspection Type	Findings	
03/09/2016	Lam Lab, Otters Lab	Quarterly Laboratory Self-Inspection	1	View
03/04/2016	Otters Lab	Quarterly Laboratory Self-Inspection	1	View
03/04/2016	Otters Lab	Quarterly Laboratory Self-Inspection	1	View
03/04/2016	Otters Lab	Quarterly Laboratory Self-Inspection	3	View
01/26/2016	Otters Lab	Quarterly Laboratory Self-Inspection	1	View
11/12/2015	Otters Lab	Quarterly Laboratory Self-Inspection	2	View
05/12/2015	Otters Lab	Quarterly Laboratory Self-Inspection	0	View

3 **Findings found during self inspections in the last 18 months**

4 **Add a new Self Inspection**

Findings Found	Category	Frequency
Areas around fire extinguishers, pull alarms, emergency eyewash showers, or electrical panels were obstructed.	Quarterly Laboratory Self-Inspection	5
18-inch vertical clearance was not maintained around fire sprinkler heads.	Quarterly Laboratory Self-Inspection	2
Equipment or furniture taller than 4 feet tall was not secured.	Quarterly Laboratory Self-Inspection	1
Life Safety Box contents were not up to date.	Lab Emergency Planning	1

B. Add a New Self inspection

Upon clicking the “**Add a new Self Inspection**” link on the **Self Inspection Log** page, the **Add NewSelf Inspection** page will appear.

1. Click on the **Inspection Type** field to select an inspection type.
2. Identify the laboratory and spaces to inspect under **Inspected Groups**.
3. The **Lookup** feature on this screenshot is not applicable to Stanford’s self inspection process. Please ignore.
4. Click on the **Inspect** button to start self inspection.

The screenshot shows the 'Add New Self Inspection' page on the Stanford University Sandbox. The page has a red header with the Stanford University logo and user information: 'Welcome, Lilly Otters', 'Home | Support | Logout', and the 'SANDBOX' logo. A left sidebar contains navigation links: 'Otters Lab' (with sub-links: View Lab Profile, Compliance Dashboard, Manage Members, Send Lab Message, Self Inspections, Manage Lab Forms), 'Research Tools', and 'My Account'. The main content area is titled 'Add New Self Inspection' and contains the following fields and buttons:

- Inspection Date:** 04/13/2016
- Inspection Type:** A dropdown menu with 'Choose...' selected. A blue arrow labeled '1' points to this field.
- Inspected Groups:** A section titled 'Please select the groups that you are inspecting.' containing three checked items: 'Otters Lab', 'ENVIRONMENTAL SAFETY FACILITY - 1100A', and 'ENVIRONMENTAL SAFETY FACILITY - 1110B'. A blue arrow labeled '2' points to the second item.
- Lookup:** A text input field with an 'Add' button next to it. A blue arrow labeled '3' points to this field.
- Inspect:** A button at the bottom of the form. A blue arrow labeled '4' points to this button.

Clicking the **Inspect** button on the previous page will bring up the **Submit Self Inspection** page.

1. If inspection is for a specific area only, enter the area in the **Inspected Area** field.
2. On the **Findings Found** section, answer “**Yes**”, “**No**” or “**N/A**” for each question.
3. To bring up a text box to enter comments or notes, click on the **Add Details** link.

Submit Self Inspection

Inspection Date:

04/13/2016

Format: 04/13/2016

Inspected Groups

Please select the groups that you are inspecting.

- ☒ Others Lab
- ☒ ENVIRONMENTAL SAFETY FACILITY - 1100A
- ☐ ENVIRONMENTAL SAFETY FACILITY - 1110B

Lookup

Add inspected area
if applicable

Inspected Area:

If inspection was targeted to a specific area, enter details here.

Inspection Type

Inspection Type:*

Quarterly Laboratory Self-Inspection

Summary:

INSTRUCTIONS

All laboratory spaces containing hazardous materials must be inspected at least quarterly.

Findings Found

Select findings to populate the Corrective Actions section with default text. This text can be edited prior to submitting the Inspection Report.

Quarterly Laboratory Self-Inspection

1. Areas around fire extinguishers, pull alarms, and emergency eyewashes/showers clear and accessible? *

Yes No N/A

Add Details

2. All objects stored at least 18 inches away from fire sprinklers? *

Yes No N/A

Add Details

3. Cabinets, furniture, and equipment taller than 4 feet braced or anchored? *

Yes No N/A

Add Details

4. Answering “**No**” to any question will trigger the **Comments/Corrective Actions** box to appear with recommended corrective actions. In this box, enter any comments and enter the date that the issues have been corrected.
5. To enlarge the **Comments/Corrective Actions** box in order to view all texts, drag the handle at the bottom right corner (where the arrow is pointing on the screen capture below).
6. To indicate specific **spaces** where issue is found, place checkmark(s) on the specific space(s).
7. To add attachments such as a work order, click on the **Add Attachments** link.

▼ **Findings Found**

Select findings to populate the Corrective Actions section with default text. This text can be edited prior to submitting the Inspection Report.

▼ **Quarterly Laboratory Self-Inspection**

	Yes	No	N/A	
1. Areas around fire extinguishers, pull alarms, and emergency eyewashes/showers clear and accessible? ⓘ*	☐	☒	☐	Hide Details
Comments/Corrective Actions: Corrective Action(s): Remove obstructions. Date Corrected:				
Space(s): ☒ ENVIRONMENTAL SAFETY FACILITY - 1100A ☐ ENVIRONMENTAL SAFETY FACILITY - 1110B				
Add Attachment(s)				
2. All objects stored at least 18 inches away from fire sprinklers? ⓘ*	☐	☐	☐	Add Details
3. Cabinets, furniture, and equipment taller than 4 feet braced or anchored? ⓘ*	☐	☐	☐	Add Details

8. Enter comments or observations that do not pertain to individual findings into the **OtherComments/Corrective Actions** box.
9. Add supporting files or upload photos in the **Attachments** section.
10. Click the **Save** button to save the inspection.

maps, and chemical inventories? ⓘ*

26. Compressed gas cylinders are listed on the chemical inventory in ChemTracker, positioned so the label is visible, and stored in a dry, well-ventilated location protected from heat sources? ⓘ*

☐ ☐ ☐ [Add Details](#)

27. Cylinders over 26 inches tall secured to a rigid structure at 1/3 and 2/3 height with metal chains, and a maximum of 2 cylinders used per pair of chains (one restraint for cylinders ≤26 inches and dewars)? ⓘ*

☐ ☐ ☐ [Add Details](#)

28. Cylinder valves closed and valve caps in place when cylinders not in use? ⓘ*

☐ ☐ ☐ [Add Details](#)

► **Other Findings**

Other Comments/Corrective Actions:

8

▼ **Attachments**

Attach new file:

No file selected.

10

LAB COMPLIANCE CHEAT SHEET

While not inclusive, the following items are lab violations commonly cited by Santa Clara County during hazardous materials inspections. **For questions, please call us at (650) 723-0448.**

HAZARDOUS CHEMICAL WASTE DISPOSAL

Drain disposal	- If you are unsure whether a material is drain disposable or trash disposable, check https://ehs.stanford.edu/reference/list-non-hazardous-chemical-wastes or contact EH&S at https://ehs.stanford.edu/contact - If the material is poured down the drain and is later determined to be hazardous, that is <u>illegal disposal</u>. Don't do it!
Waste label and disposal	- Waste tags must be attached to hazardous chemical waste containers when the first drop of waste is collected. - Waste tags are created using the Online Waste Tag System at http://wastetag.stanford.edu - Waste containers must be kept closed except when adding waste. - Funnels must be removed after use and the waste container immediately recapped. - Submit a waste pickup request when container is full or if it has been eight months past the accumulation date at http://wastetag.stanford.edu. - For waste tags not generated using the Online Chemical Waste Manager, submit an online pickup request through http://wastepickup.stanford.edu
Spill debris	- Rags and towels used to clean up hazardous material spills, including vacuum pump oil, become hazardous waste and must be disposed of as such. Do <u>not</u> throw into the regular trash. - On a waste tag, these can be described as "solids contaminated with (chemical name)".
Empty container management	- Empty Containers Disposal: - Use the "three shakes" rule to determine if a container is completely empty. - Invert the container over an appropriate hazardous waste container, shake three times, and wait several seconds after the last drop is seen. For solids and viscous materials, scrape the container to remove any residue. - Once the container is empty, it can be tossed into the trash or recycling bin UNLESS it is an "extremely hazardous waste" or an "acutely hazardous waste" https://ehs.stanford.edu/reference/list-extremely-hazardous-wastes https://ehs.stanford.edu/reference/list-acutely-hazardous-wastes https://ehs.stanford.edu/forms-tools/empty-container-decision-tree - No evaporation of waste!
Non-contaminated broken glass	Check that non-contaminated broken glass collection boxes only contain glass. Make sure that there are no liquids in bottles, gloves, or trash.
Solder waste	Solder waste should be treated as hazardous waste. See: https://ehs.stanford.edu/reference/soldering-safety-reference-sheet

CHEMICAL STORAGE AND LABELING

Secondary container	- All hazardous chemicals must be stored in secondary containment and segregated according to chemical compatibility. - Example: Containers of bleach (storage group E), ammonia (storage group C), and ethanol (storage group L) must be in separate secondary containment.
Container type	- All chemical containers must be structurally sound and tightly capped. - Leaks into secondary containment must be cleaned up immediately (example: standing oil from a leaking vacuum pump).
Labels	- All containers must be labeled with their full chemical name in English (example: label “methanol” and not “MeOH”). - Exceptions to this rule: - If the container holds a nonhazardous buffer with pH 5.5-11, you may use the buffer abbreviation (example: PBS, TRIS, TBS). - If the contents are a manufactured product with a trade name, you may use the trade name (example: Zaclon ZR flux). - If you plan to reuse a chemical, label the container “for reuse” next to the full chemical name (example: acetone – for reuse). Do not use the labels “Dirty” or “Used”.
Flammable storage	- Store flammable liquids in flammable cabinets in secondary containment. - In some areas, the California Fire Code permits storage of small quantities of flammable liquids outside flammable cabinets. - Contact the Stanford University Fire Marshal’s Office for additional guidance.

GASES

ChemTracker	- All compressed gas cylinders and liquefied gas dewars must be listed on the online chemical inventory system (ChemTracker) for the room in which they are stored. - The reported amount must be the size of the gas cylinder or dewar.
Restraint	- Gas cylinders >26” tall must be restrained by metal chains at 1/3 and 2/3 of the cylinder height. - A maximum of two cylinders can be restrained using the same set of chains. - This applies to in-use, stored, and empty cylinders. - Dewars and cryotanks must be restrained at a minimum of one point. - This applies to in-use, stored, and empty dewars and cryotanks.

INSPECTION RECORDS

Quarterly self-inspections	Make sure you have at least three years’ worth of quarterly self-inspection records either uploaded to BioRAFT or hard copies.
Monthly hazardous storage inspections	Make sure you have at least three years’ worth of monthly hazardous storage inspections either uploaded to BioRAFT or hard copies.

EMERGENCY EYEWASHES & SAFETY SHOWERS

Access	- Do not store boxes or equipment adjacent to or under eyewashes and safety showers. - This presents a danger to all lab personnel should anyone need to access the eyewash or safety shower during an emergency.
Testing and maintenance	- For Stanford University laboratories, the Stanford Plumbing Shop tests all campus eyewashes and safety showers monthly. - For Stanford School of Medicine laboratories, the Facility Operations group in the Office of Facilities Planning and Management (OFPM) tests eyewashes and safety showers.

REFRIGERATORS & FREEZERS

Storage	- Flammable and combustible materials must only be stored in refrigerators and freezers rated for flammable storage. - Household refrigerators and freezers are not designed and are not approved for this type of storage. - Incompatible hazardous materials stored in refrigerators and freezers, including in their doors, must be segregated and secondarily contained according to their storage groups.
---------	--

ETHIDIUM BROMIDE AND GELS

Ethidium bromide waste	- If the ethidium bromide solution you use is over 4 mg/ml (0.4 weight percent), it must be handled as hazardous waste. - Know the concentration of the ethidium bromide you work with; you may be asked this by an inspector.
Gels	If a gel has less than 0.4 weight percent of ethidium bromide, it can be disposed of in the trash.

UNIVERSAL WASTE (BATTERIES, LAMPS, E-WASTE)

Fluorescent and UV lamps	- Fluorescent and UV lamps are considered universal waste and cannot be thrown into regular trash or broken glass boxes. Contact your building's facilities coordinator for disposal. - Place in a hard-sided container, label as "universal waste," identify what the contents are, and indicate the date that you are designating it as waste. - The lamps must be removed from labs no more than nine months past the date you first designated them as universal waste.
Labels	Universal waste labels may be obtained from EH&S.

Medical Waste Common Inspection Findings

While not all-inclusive, the following items are commonly cited by Santa Clara County during medical and biohazardous waste inspections. **For questions, call EH&S at (650) 723-0448.**

Containment and Storage of Biohazardous Waste	
Bagged biohazardous waste must be in red bags. Do <u>not</u> use orange, pink, or clear autoclave bags. Biohazard bags must meet both ASTM tests: ASTM D1922 tear resistance and ASTM D1709 impact resistance.	
Bagged waste must be stored, handled, and transported in proper containment: • Rigid hard-sided container • Leak resistant and have tight-fitting covers • Kept clean and in good repair • Labeled on the lid and sides so as to be visible from any lateral direction - must use the words "Biohazardous Waste" or the international biohazard symbol and the word "BIOHAZARD"	

Open and unlidded biohazardous waste bins

Do not leave the lids open and unattended.

Biohazardous waste bins must remain closed to maintain a sanitary working environment and to prevent spillage of contents in the event of an emergency.

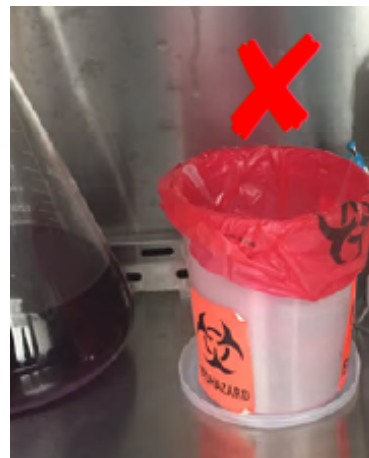
Remember to keep all biohazardous waste bins closed when not in use.

- Close or replace lid before you walk away
- Do not overfill bins
- Request a waste pick-up when bins are over 2/3 full



Benchtop biohazardous waste bins in labs and tissue culture rooms must be closed when unattended.

- As a best practice, tie the bag and place it into biohazardous waste bin when your work is complete



Improper Sharps Disposal

“Sharps waste” are items that have acute rigid corners, edges, or protuberances capable of cutting or piercing, including but not limited to:

- Hypodermic needles with syringes, needles with attached tubing, blades and scalpels, broken glass, Pasteur pipettes

Place contaminated sharps into approved sharps containers only:

- Rigid, puncture-resistant containers with the words “sharps waste” or with the word “BIOHAZARD”



Do not store full sharps containers for more than 30 days.

- Place any broken/damaged, faded, or outdated sharps containers into the biohazardous waste collection bins for immediate pick up



Open Sharps Containers

Sharps containers without safety trays must remain closed to maintain a sanitary working environment and to prevent spillage of contents in the event of an emergency.

- As a best practice, remember to keep sharps containers closed when not in use
- Close or replace lid before you walk away
- Do not overfill bins and containers
- Request a waste pick-up when bins are over 2/3 full
- Lock filled sharps bins when ready for pick-up



LAB LEVEL MEDICAL WASTE INSPECTION GUIDANCE

[BIOSAFETY MANUAL](#) - Chapter 11 for more detailed information

Biohazard Bags as primary container (solid waste)

- Biohazard bags used for solid non-sharps waste must be red in color, labeled with the word “BIOHAZARD” and the universal biohazard symbol (No orange, clear, or black bags).
- Biohazard bags must meet both ASTM tests (ASTM D1922) tear resistance and (ASTM D1709) impact resistance tests.
- When finished adding to biohazardous bags, they shall be tied to prevent leakage or expulsion of contents during all future handling and storage (i.e., knot, gooseneck, tie or tape).

Biohazard Waste Containers as secondary container (solid waste)

- Biohazardous waste bags containing waste must be enclosed within a solid waste closed container that is rigid, puncture resistant, leak resistant, composed of a smooth cleanable material, properly labeled with universal biohazard symbols and “BIOHAZARD” on all visible sides:
 - rectangular container must have symbols on all 4 sides and the lid
 - circular container must have symbols on all equilateral points and the lid
- **Tight fitting lids must be closed on all biohazardous waste containers. Lids must be closed when you are not actively collecting waste.**
- No items should ever be placed on top of the medical waste containers, so that the biohazard symbols are viewable at all times.
- Red biohazard bags containing medical waste must be shipped off site within 7 days or 30 days if stored below 32 degrees fahrenheit.
- Biohazard waste containers must be stored in the lab where they are generated or moved directly to a designated accumulation area (DAA). Designated Accumulation Areas must be secure and locked rooms designated for the storage of medical waste before it is shipped off site by the vendor. Biohazardous waste must not be unattended or left in public hallways, autoclave rooms, or left outside of a designated accumulation room.

Liquid Biohazardous Waste

- Dilute with appropriate disinfectant or bleach 1:10 (final concentration = 0.5% sodium hypochlorite), allow 20 minutes for contact time, then dispose into the sewer with running water.

Sharps Waste

- Any waste with acute rigid corners, edges, or protuberances capable of cutting or piercing human skin (i.e., needles, glass pipettes) must be discarded in a sharps waste container.
- DO NOT OVERFILL sharps containers. Properly dispose of them when they are full.
- Add sharps only to the “Fill to Here” line; cover and dispose of containers when they are $\frac{3}{4}$ full.

- Sharps containers must have proper fitting lids that are closed when items are not being added and once $\frac{3}{4}$ full secure the lockable lid feature. Sharps containers without safety trays must be kept closed when not in use.
- Non-biohazardous sharps containers are clearly marked as “non-contaminated” and all biohazard markings are removed, covered or defaced.
- Pharmaceutical wastes are not discarded in sharps containers, but in a pharmaceutical waste container. Contact EH&S if you are unsure if your waste is pharmaceutical waste.

Medical Waste Transportation on Campus

- Prior to transport to a designated accumulation area (DAA), bags are secured and closed inside the lab and then transported to the DAA inside properly labeled, rigid, secondary containers. The transport container lid must be secured shut and the exterior of the container must be decontaminated before leaving the lab.
- All medical waste is transported in a clean, closed secondary container on a cart.
- The secondary transport container must be rigid, puncture-proof, leak resistant, and be labeled with the biohazard symbol and the word “Biohazard” on the lids and all lateral sides of the container.

Training

- If the researcher is generating biohazardous and medical waste containing bloodborne pathogens or other potentially infectious materials, then Bloodborne Pathogens EHS-PROG-1600 must be completed annually.

Segregation and Minimization of Medical Waste

- Non-contaminated paper towels, non-contaminated gloves and packaging materials should be disposed of as regular trash and not biohazardous waste.

Carcass Disposal for Labs Working with Animals

- Carcasses should be disposed of in a black plastic bag that is labeled with the protocol number.
- Red biohazard bags should ONLY be used to dispose of carcasses that have been exposed to BSL-2 or 3 agents.
- Red biohazard bags must never be used to carry or drape rodent cages during transport.

[Roles](#)
[Health & Safety Topics](#)
[Services](#)
[Training](#)
[Guides & Manuals](#)
[Standard Operating Procedures](#)
[Forms & Tools](#)
[EH&S News](#)

Safety Store

TOPICS

- ☐ Biological Safety Supplies (17)
- ☐ PPE (6)
- ☐ Chemical Safety Supplies (18)
- ☐ Complimentary (36)
- ☐ Documents and Manuals (4)
- ☐ Emergency and First Aid Supplies (5)
- ☐ External Vendors (27)
- ☐ General Safety Supplies (8)
- ☐ Posters, Stickers, and Signage (28)
- ☐ Radiation Safety Supplies (12)
- ☐ Waste Disposal (11)

Welcome to Stanford University's EH&S's Safety Store! This is an online store where we have compiled a variety of equipment, resources, and supplies to buy, use, or borrow at your convenience. Many of the items available on the store are complimentary from our department, while others are linked to external vendors for purchase. Please browse our site, and we hope that your research and academic endeavors are made easier with this one-stop shop!

**Disclaimer: Ensure you have appropriate safety training from your local supervisor before purchasing and/or using the supplies on this site.*

Showing 1–16 of 66 results



**5 Gallon Poly Pail for
Lab Debris**



**2.5 Gallon HDPE
Carboys**



Waste Tag Sleeves



Safety Glasses



Lab Coat Vendor



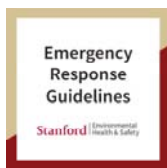
**Non-disposable
Gloves**



Cut Resistant Gloves



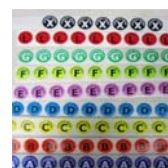
First Aid Kits



**Emergency
Response Guidelines**



**Chemical Spill
Cleanup Kits for
Small Spills**



**Storage Group
Stickers**



**Medical and
Biohazardous Waste
Disposal Guidelines
Poster**



Geiger Counters



**4 Liter Plastic Coated
Glass Bottles with
Vented Lid**



**Secondary
Containers**



**"Disposal of Sharps
and Broken Glass"
Poster**



Wastewater: Glass Wash Standard Operating Procedures

Stanford Water Resources and Civil Infrastructure

suwater.stanford.edu

The following procedures have been developed as part of a comprehensive program to reduce the amount of pollutants that reach the sanitary sewer system and the South San Francisco Bay. This SOP provides guidance for laboratory users on how hand washing glassware can be performed while remaining in compliance with wastewater general discharge permit requirements from the Palo Alto Regional Water Quality Control Plant (RWQCP).

1. **Do not dispose of chemical waste down the sink drain, into the sewer, or into regular trash containers.** Check chemical disposal guidelines for proper handling.
2. If glassware is hand washed with acids, bases, solvents, or alcohols, triple-rinse and dispose of all three rinses as hazardous waste, along with any waste product. All rinsate should be drained thoroughly into the appropriate waste container. Do not mix incompatible wastes. To properly triple rinse, follow these steps:
 - a. Use clean tap water or DI water and add enough to the container (approximately 10-20% of its capacity), recap or seal if appropriate, swirl carefully to all sides, and empty contents into Hazardous Waste (HW) containers. Repeat two more times.
 - b. The rinsate should always be collected in an intact sealable, HW container which is compatible with the waste. Affix a completed HW label, which should be applied as soon as the first drop is generated. Go to wastetag.stanford.edu to create the tag.
 - c. The empty container or glassware should be drained so no liquid pours out of the container, or does not drip when inverted. After the third rinse is drained to the hazardous waste container, the glassware can be allowed to air dry on a drying rack (or equivalent).
3. If needed, soak glassware in base cleaning bath (e.g. potassium hydroxide)
 - a. Remove glassware from base cleaning bath, and then dip into a buffer rinse solution (e.g. 0.5 M Sodium Citrate solution).
 - b. Remove glassware from rinse buffer solution; allow it to drain thoroughly into the buffer container.
 - c. Rinse glassware with water.
 - d. Test the pH of the buffer solution and transfer it to the appropriate acid or base waste container.

If you would like more information on your building's general discharge permit, contact the building or lab manager. More information about wastewater best management practices is available at suwater.stanford.edu/laboratories.

Contact the Water Planning & Stewardship team if you have questions or concerns regarding this SOP or other wastewater compliance items. Find our contact information at suwater.stanford.edu/wps.



Wastewater: Best Management Practices Fact Sheet

Stanford Water Resources and Civil Infrastructure

suwater.stanford.edu

About the General Discharge Permit and Sewer Use Ordinance

The purpose of the General Discharge Permit is to regulate wastewater discharges from lab buildings at Stanford University (Stanford) which ultimately end up in the City of Palo Alto's sanitary sewer system. Specific buildings are regulated under this General Discharge Permit which have similar types of operations, wastes, and requirements for monitoring with effluent limits. Wastewater discharges shall comply with Chapter 16.09 of the Palo Alto Municipal Code ([Sewer Use Ordinance](#)), any applicable provisions of Federal or State laws and regulations, and all provisions of the permit itself.

Best Management Practices

As part of a comprehensive program to reduce the amount of pollutants that reach the sanitary sewer system and the South San Francisco Bay, the Palo Alto Regional Water Quality Control Plant (RWQCP) has prepared these Best Management Practices (BMPs) for laboratories and industrial facilities.

I. Requirements

A. Wastewater Discharge

- Hazardous wastes, as defined by either State or Federal regulations, including California Code of Regulations Title 22 and Code of Federal Regulations Title 40 Part 261, shall not be disposed of via the sanitary sewer. Contact Stanford's Water Planning & Stewardship team (suwater.stanford.edu/wps) to determine what substances are appropriate for discharge to the sanitary sewer under the Sewer Use Ordinance. See attached table for local discharge limits and detection levels.
- The first three rinses from lab equipment that has been in contact with hazardous or prohibited materials (e.g., chemical containers, glassware, apparatus, etc.) shall not be discharged to the sanitary sewer unless authorized in advance by Stanford Utilities and the RWQCP. This includes the first three rinses from hand washed glassware using acids, bases, solvents, or alcohols. The first three rinses should be containerized and disposed by EHS.
- Chemical reagents or additives containing metals such as thimerosal or copper containing algaecides shall not be discharged to the sanitary sewer unless authorized in advance by Stanford Utilities and the RWQCP.
- The use of water aspirators is prohibited.

B. Chemical Storage and Secondary Containment

- Use a secondary containment vessel or area (e.g., a tray, canister, or bermed area) that is impervious to liquid(s) being contained and large enough to hold at least 110% of the capacity of the primary container, or for multiple primary containers 150% of the largest primary container.
- Secondary containment vessels or areas should not drain into any sink or sewer drain and chemicals should not be stored in or above sinks.





Wastewater: Best Management Practices Fact Sheet

Stanford Water Resources and Civil Infrastructure

suwater.stanford.edu

C. Personnel Orientation and Training

Personnel, including workers and supervisors whose duties pertain in any manner to the production or removal of waste discharges, shall be informed of the provisions of the Sewer Use Ordinance. Steps to inform such personnel include:

- Orientation of newly employed or assigned personnel.
- Annual orientation of all appropriate personnel.
- Posting signs or diagrams at all workstations that indicate approved waste disposal methods and accidental leak or spill reporting requirements.



II. Additional Measures

A. Replace Mercury Containing Devices with Non-Mercury Alternatives

Devices containing mercury, including mercury thermometers, should be replaced with non-mercury alternatives.

B. Container Security

To reduce the chances of chemicals being released to the sanitary sewer or the laboratory in the event of a spill or earthquake:

- Purchase smaller quantities of chemicals. Smaller containers allow for easier storage and more efficient consumption of chemicals.
- Use plastic coated chemical bottles to help reduce bottle breakage. Use plastic or insulated holders for solvent bottles. Store glassware and containers on rubber mats.
- Secure chemicals stored on chemical safety shelves or in cabinets behind barriers (at least 1/5 the height of tallest container) if secondary containment is not feasible.
- Make sure all chemical containers are properly labeled and segregate incompatible chemicals to prevent mixing in case of a spill.
- Have copies of Safety Data Sheets (SDS), formally called MSDS, on file for all chemicals stored or used at the facility.



C. Chlorinated and Organic Solvent Compliance

All chlorinated and organic solvents must be kept completely out of the effluent water lines (drains, sinks, water recirculative pumps, aspirators, etc.)

- Glassware from solvent use must be rinsed with acetone or methanol and rinsate waste collected as hazardous waste.
- All flammable and organic solvents (flash point below 140°F and/or above 1,000 mg/L) must be collected as hazardous waste (including acetone, ethanol, methanol, isopropyl alcohol, and toluene).
- Use a special rotary evaporator to remove chlorinated solvents. The evaporator must either operate off an air cooled vacuum pump or use the house vacuum system.
- All liquid waste shall be collected and no effluent shall be disposed to sinks.
- Tubing that might contain solvents must not terminate in sinks.
- Water fractions from CH₂Cl₂ (Methylene Chloride) extractions contain 20,000 times the sewer limit (0.75 mg/L) and should be collected.



Wastewater: Best Management Practices Fact Sheet

Stanford Water Resources and Civil Infrastructure

suwater.stanford.edu

D. Solder Disposal

The California Department of Toxic Substances Control (DTSC) has determined that processed solder residues from soldering operations contain metal oxides which exclude them from the scrap metal exemption found in 22 CCR §66260.10. Therefore, all solder residues must be managed as hazardous waste, including lead-free solder.



- Use re-closeable storage bags (such as Ziploc bags) for containing solder waste by taping them on or near soldering workstations. These bags must be kept closed except when adding solder waste. Attach a label to the bag once it is used to hold solder waste.
- Submit a pick-up request at no later than 8 months past the accumulation date. Use the Stanford on-line system for creating waste labels and requesting pickup: wastetag.stanford.edu.
- Dispose of all solder/flux contaminated wipes, sponges and debris as hazardous waste.
- Use solder made from less hazardous materials (i.e. silver and nickel) whenever possible.

E. Waste Minimization

Basic waste minimization techniques include micro-scale experiments, product substitution, reduced product usage and storage, recycling/reuse of chemicals, modified operations, and water conservation.

- Substitute chemicals with less toxic alternatives.
- Use the minimum amounts of chemicals required by each experiment or process to minimize disposal volume at end of procedure.
- Order minimum amounts of chemicals to reduce waste and leftover materials in case procedures are changed or expiration dates pass.



F. Experiments and Equipment

Many types of equipment used routinely in conducting experiments may generate wastes that should not be discharged to the sanitary sewer.

- Use a rotary evaporator with a contained vacuum system and a dry ice condenser to intercept solvents.
- Drains from analytical equipment and automated chemistry should drain to waste containers.

G. Water Conservation

Structural measures such as those listed below can reduce water use significantly. Minimize water use where practical. For example:



- Install water-saving devices (such as flow restrictors) on sinks and rinse tanks (see photo on right).
- Reduce rinse times if possible (without affecting product quality).
- Eliminate the use of potable water as a single pass coolant. For restrictions on the use of single pass cooling, refer to Section 16.09.055 of the Sewer Use Ordinance.



For more information on your building's general discharge permit, contact the building or lab manager.

Contact the Water Planning & Stewardship team if you have any questions or concerns regarding these laboratory BMPs or other wastewater compliance items: suwater.stanford.edu/wps

Or visit our website for more information: suwater.stanford.edu/laboratories



Wastewater: Best Management Practices Fact Sheet

Stanford Water Resources and Civil Infrastructure

suwater.stanford.edu

Local Discharge Limits and Analytical Detection Levels

Pollutants	Local Maximum Limits ¹ (mg/L)	Maximum Allowable Analytical Detection Levels (mg/L)
Arsenic	0.1	0.01
Barium	5	0.5
Beryllium	0.75	0.075
Boron	1	0.1
Cadmium	0.1	0.01
Chromium, Hexavalent	1	0.1
Chromium, total	2	0.2
Cobalt	1	0.1
Copper	0.25 ²	0.025
Cyanide	0.5	0.05
Dissolved Sulfides	0.1	0.01
Fluoride	65	6.5
Formaldehyde	5	0.5
Lead	0.5	0.05
Manganese	1	0.1
Mercaptans	0.1	0.01
Mercury	0.01	0.001
Methyl Tertiary Butyl Ether (MTBE)	0.75	0.075
Nickel	0.5	0.05
Phenols	1	0.1
Selenium	1	0.1
Silver	0.25	0.025
Single Toxic Organic	0.75	0.075
Total Toxic Organics	1	0.1
Zinc	2.0 ³	0.2
pH	5.0 pH units	11.0 pH units

¹ For discharges with annual average flows greater than fifty thousand gallons per day through any single sampling location, the maximum allowable limits shall be one-half the values listed in the table, with the exception of copper, mercury, MTBE, nickel, and silver, for which the limits shall remain 0.25 mg/L, 0.010 mg/L, 0.75 mg/L, 0.50 mg/L, and 0.25 mg/L, respectively, regardless of flow.

² The local maximum copper limit for cooling system discharges less than 2,000 gpd, Vehicle Services, Photoprocessing, Machine Shops shall be 2.0 mg/L. See Section 16.09.045 of the Sewer Use Ordinance for details and for metal finisher requirements.

³ The local maximum zinc limit for vehicle service facilities shall be 4.0 mg/L

Shared Lab Spaces – EH&S Guidance

Shared Lab Spaces: Shared lab spaces, where lab equipment and/or hazardous materials are used/stored, are located throughout university lab buildings. Examples of shared lab spaces include:

- fume hood alcoves in open lab suites used by one or more Principal Investigators (PIs)
- analytical equipment located in a PI's laboratory shared by others
- core facilities used by a department
- teaching areas

Challenges: As shared lab spaces are utilized by multiple individuals from different labs, proper safety management of lab equipment, supplies, hazardous materials, and housekeeping can be challenging. Common problems include: contamination that is not promptly cleaned up, hazardous waste that is not properly labeled, or poor general housekeeping. Without proper management of shared spaces, potential safety and compliance issues may arise.

Adhere to the following guidelines for proper management of shared spaces:

Hazardous Materials Use:

- Ensure all stock and in-use containers of chemical, biological, and radioactive material are appropriately labeled.
- Clearly identify work areas where experiments are in progress; label with appropriate hazards, research group, contact name(s), and contact phone number(s).
- Ensure all hazardous waste containers are appropriately labeled with hazardous waste tag (wastetag.stanford.edu).
- Request hazardous waste pickup for hazardous waste accumulation containers older than 8 months (wastepickup.stanford.edu).

Personal Protective Equipment (PPE):

- Ensure a [Laboratory PPE Assessment](#) has been completed for the work area and applicable procedures.
- Wear appropriate PPE to match the potential hazards present based on each PI's Laboratory PPE Assessment.
- Update PPE assessment when new hazards are introduced, PPE requirements change, or as needed.

Equipment:

- Label all equipment with owner's name and contact information.
- Obtain lab-specific (Tier III) training prior to use of shared lab equipment. Training shall be provided by equipment owner or his/her designee.

Work Practices:

- Clean work area at the end of operation or the end of the day.
- Clean out shared space periodically (e.g., every two weeks).
- Keep fume hood(s) clear of clutter.
- Perform [quarterly self-inspections](#) of entire shared space.
- Inventory all hazardous materials stored in space by your group via ChemTracker (stanford.chemtracker.org).
- Communicate health and safety concerns to other members of the shared space.
- Report health and safety concerns to your PI.

Responsible Parties:

- Identify specific faculty member or lead PI (if multiple PIs occupy the shared space) that all personnel can approach with safety concerns if not resolved by local conversations.
- Designate personnel (per lab group) to update chemical inventory, conduct periodic lab clean-outs, perform [quarterly self-inspections](#) and communicate issues back to home lab group.
- Update Life Safety Box contact sheet annually or when designated personnel change.

Emergency Response:

- **Health threatening:** Call 9-911 from campus telephones (286 from SoM), alert others, and evacuate.
- **Non-health threatening:** Clean up spills immediately. Call 725-9999 for spill assistance

For further support on shared lab space guidance, contact Environmental Health and Safety (EH&S) at 723-0448.

CHEMICAL INVENTORY & MANAGEMENT

Chemical Inventory Management

<https://ehs.stanford.edu/topic/chemical-safety/chemical-inventory-management>

Chemical Labeling

Label all chemical containers with the identity of the contents.

- List the full chemical name(s) in English
- Do not use chemical formulas, chemical structures, abbreviations or acronyms
 - Exceptions to this rule
 - If the container holds a nonhazardous buffer with pH 5.5-11, you may use the buffer abbreviation (e.g., PBS, TRIS, TBS)
 - If the contents are a manufactured product with a trade name, you may use the trade name (e.g., Zaclon ZR flux, windex)
- Include hazard warning (e.g., flammable, corrosive, toxic, reactive)
- Include chemical concentration information
- Hazardous Waste Containers must be labeled with a waste tag

Chemical Inventory

Federal, state, and local regulations require that Stanford keep an accurate inventory of the chemicals stored on campus. We report this chemical inventory to the governing regulatory agencies in the form of the Hazardous Materials Management Plan (HMMP). The Plan also helps users keep track of their chemicals, conduct laboratory hazard assessments, provide chemical hazard information to emergency responders, and minimize unnecessary stockpiling of chemicals.

How do labs manage their inventory?

PIs and lab or shop managers must report the hazardous materials they have on hand via ChemTracker 4 online application to inventory all the hazardous chemicals on campus. Each lab and shop that uses chemicals must update their chemical inventories in ChemTracker 4 at least annually.

EH&S ChemTracker 4 information: <https://ehs.stanford.edu/services/chemtracker>

Request new ChemTracker 4 user account: <https://ehs.stanford.edu/request-a-new-chemtracker-user-account>
ChemTracker 4 Demos/User

Guides: https://ehsapps.stanford.edu/chemtracker/help/inventory_management.html

What needs to be inventoried in ChemTracker 4?

What needs to be included:

- Containers on-site for more than 30 days
- It is important to pick the correct physical state, so that it will link with the chemical database
 - Liquid, Solid or Gas
- Any amount of
 - Chemicals with acute toxicity
 - Chemicals listed as a carcinogen or reproductive hazard
 - Chemicals that are pyrophoric, water reactive or potentially explosive
 - Compressed gases (including liquefied gases) – list each gas and the concentration
 - For compressed gases, if a hazardous material or mixture is determined to exceed threshold quantities at standard temperature and pressure, it shall be reported in the physical state at which it is stored. (For example, propane and liquefied nitrogen = report in gallons). For gases at Standard Temperature and Pressure report in cubic feet.

- Containers >100 mL of a liquid or >250 grams of a solid substance
 - Flammable/combustible liquids (e.g. alcohols, solvents, lubricants, and paints)
 - Flammable solids (e.g. magnesium, sodium, organo-metallics)
 - Oxidizers (e.g. concentrated mineral acids, bromates, chlorates, and permanganates)
 - Organic peroxides (e.g. hydrogen peroxide (>8%) and benzoyl peroxide)
 - Poisons not included in the above categories (e.g. biological toxins, select agent toxins, dyes, adhesives, and specimen solutions)
 - Corrosives (acids or bases that are corrosive to the skin, and other materials having a pH of ≤ 5.5 or ≥ 11.0) – provide the concentration
- 55 gallons of hazardous waste must be included in ChemTracker 4
 - Enter – Waste (chemical name)
- Hazardous Waste that is shipped directly from facility (offsite locations). EH&S is responsible for inputting this data into Chemtracker 4
- [NSAR Select Toxins](#)
- [Controlled Substances](#) (DEA-listed drugs)

It is not necessary to inventory in ChemTracker 4:

- Retail products used for routine household-like activities (e.g. cleansers and dish soap). However, do include bleach used in laboratory processes.
- Materials that will be expended within one or two days (e.g. working solutions)
- Radioactive materials
- Biohazardous materials
- Non-hazardous buffers
- Growth media
- Enzyme preparations
- Sugars

ChemTracker 4 support

EH&S provides customer support for ChemTracker 4 at Stanford. This includes: adding new chemical owner accounts (i.e., Principal Investigators), new ChemTracker 4 user access, and providing training for new users and basic troubleshooting.

You can contact the HMMP group for the following issues:

- [Request new ChemTracker 4 user account](#)
- [Request new chemical owner account \(for the responsible PI or supervisor\)](#)
- [Request life safety box printout](#)
- [Other requests](#)

For ChemTracker 4 user basics, check out the quick guides:

[Part 1: Login/Logout, Layout, ChemInfo & MSDS \(rev. 10/28/13\)](#)

[Part 2: Add Inventory, Add Inventory Templates](#)

[Part 3: Search, View, Download, Saved Search Templates](#)

[Part 4: Modify, Surplus](#)

For additional questions please contact EH&S at (650) 723-0448.

Modify Inventory from Details View

Make updates to records after a Search to locate them. View the records in the *Details View*.

Delete

Select contiguous or non-contiguous record(s) and press *Delete* at the bottom.

Duplicate

If adding another container of an item already in inventory, search for the existing record, go to the Details View, select the record, and press *Duplicate*. You can duplicate more than one record at a time. Certain fields are not duplicated, such as Barcode and Expiration Date, as well as system-generated fields such as Created By.

Changing Values in a Single Record

1. Select a single record and press the *Modify* button. Cells indicate whether they may be edited.
2. Enter the new value(s).
3. Press *Save* in the far left *Actions* column.

Bulk Edit (changes to multiple records)

Use this feature to make the same change(s) to multiple inventory items simultaneously, e.g., moving inventory to a new owner and/or location, or for storeroom receiving.

1. Select multiple items (do not have to be contiguous) and press *Modify* or *Modify all* at the bottom.
2. On the Bulk Edit screen, make your changes and press *Save Modified Records*.



Add Inventory

1. Press *Add* in the top navigation bar.
2. On the *Add Inventory* screen, complete the 9 required fields at the top (Owner, Department, Building, Room, Chemical Name, Physical State, Container Count, Amount, Units).
3. Use pull-down menus when you can to:
 - Help you enter validated data
 - Link your item to the reference database (using certain fields)
 - Auto-populate data from the reference database into other fields (in some cases).
4. Enter information into other fields as appropriate.
5. Press *Add Inventory* at the bottom.

TIPS

- Try to enter the Manufacturer and Product Number first. If the combination is in the reference database, this may be the fastest way to get chemical information entered since other data will be auto-filled for you.
- Create templates for often-used Owner, Department, Building and Room information.
- Don't add a CAS Number when adding a mixture.

For more information:

1. Click the *Help* button in ChemTracker
2. Contact the CT Administrator at your site

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*The higher education consortium for
chemical inventory management and
reporting*

Using CT4 Inventory Management Quick Start

This brochure is designed to give you the basic information you need to manage your chemical inventory using ChemTracker 4 (CT4). **Changes from previous versions of CT are marked with an asterisk (*); important points are in bold.**

Webinar training videos, browser settings, detailed instructions, screen shots, and examples can be found by pressing the *Help* button from within the application.

To **login**, go to chemtracker.stanford.edu and click *Members Access Links* or ask your ChemTracker Administrator for the link. Then select the CT4 link for your institution. Enter your ChemTracker Username and Password, which are case-sensitive.

Logout under your UserID displayed in the upper right of the screen.

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General Information

- **Forgotten Usernames and Passwords**, as well as resetting Passwords, are handled by the local CT Administrator.
- **Barcode Data Entry and Searching** is available to assist with auditing.
- **CT Administrative functions and regulatory reporting** are performed in v1.

Navigation

- **Function buttons are at the top and window-specific-action buttons are at the bottom of the screen.**
- **Row Actions are in the left column.**
- **Browser Back and Forward buttons can be used to navigate.**
- To **Presearch a field value**, type in the beginning characters and view the pull-down list.
- To **clear fields**, use the *Clear selections* button, backspace over the text, or pick the first (empty) value at the top of a pull-down list.

ChemInfo is available to all CT Users, even those without specific authority to access inventory.

ChemInfo & SDSs

To see **ChemInfo**, select *ChemInfo* at the top of the screen and search, or select the *ChemInfo* Action at the left of an inventory item in the Summary or Detail View. ChemTracker displays validated physical properties, physical hazards, health hazards, storage codes, regulations and other identifiers from its reference database of over 36,000 chemicals and 110,000 associated synonyms.

Following the steps as above, select *SDS* to view a third-party SDS of the chemical(s) if your organization has a subscription.

Searching for Inventory

1. Press *Search* on the top navigation bar.
2. ***Fields are organized by expandable and collapsible categories.**
3. Specify search values for the fields. Fields you can search on include:

Chemical Name	CAS number
Formula	Barcode number
Manufacturer	Product number
Storage Group	Building/Room
Owner	Department
Modified date	Expiration date
Regulation	User-defined fields
Hazard(s)	Location
Received Date	
4. **To find inventory by Hazard(s), Regulations or Storage Group, the inventory item(s) must be linked to the reference database.**
5. You can refine your search and narrow your results by entering more field values.
6. ***You can search by multiple hazards** by entering the first letter of hazards repeatedly after you select a value from the presearch box.
7. ***You can search by Regulation** by selecting the Regulation Type, Class and ID.
8. Perform the search by clicking *Search* at the bottom of the screen. Results will match ALL criteria selected. The first 500 results are displayed in the *Summary View*. Scroll down to see 250 additional values with each scroll.

Linking

- It is important that items in inventory are linked to CT's reference database containing the safety and regulatory information.
- Items are linked automatically upon entry or update when Chemical Name + Physical State, or CAS + Physical State, or Manufacturer + Product Code match the database's and don't conflict with each other. Linked items display a number in the GDN field.
- Items can also be linked manually in a process called SAMONSing performed in v1.

Summary and Detail Views

1. ***Storage Group** is among the fields shown in the Summary View.
2. ***For linked items, the Summary View shows the Main Name of chemicals as they are in the CT database so the records can be summarized. The Details View shows the Chemical Name as it was entered into CT.**
3. ***Summary View** displays metric units. In the *Details View*, you see units as entered, either English or metric. You can select between metric and English units for display in either View using the *Change units* button.
4. **Sort** columns by pressing the small triangles to the right of the column header.
5. ***Move** columns by clicking and dragging the column header to another location.
6. Press **Hide/Show Columns** at the bottom to change the columns you see displayed.
7. ***Select one or all rows in Summary View to Download to Excel on a Mac or pc. In Details View, you can select any number of contiguous or non-contiguous rows to Download.** If you do not have a version of Excel that accepts .xlsx, go to Desktop Setup on the *Help* page.

Templates

- Create Saved Search Templates and Add Inventory templates: Give them titles and descriptions.
- Templates are executable from the home page and the appropriate function page.
- ***You can Load, Execute, and Delete Saved Searches.**
- **Tip:** Create a template, load it and change the room (or other fields) to make one template easily function as many!

STANFORD COMPATIBLE STORAGE GROUP GUIDE

Effective segregation in chemical storage reduces the risk of dangerous chemical reactions.

This guide must be used in conjunction with information from the manufacturer's safety data sheets and chemical-specific expert knowledge.

This storage group system is intended to be used in research settings to store laboratory-scale quantities of chemicals.

What to Segregate



Compatible Organic Bases



Compatible Pyrophoric & Water-Reactive Materials^{*} 



Compatible Inorganic Bases



Compatible Organic Acids



Compatible Oxidizers & Peroxides^{*}
(not including Strong, Oxidizing Acids)



Compatible Inorganic Acids
(not including Oxidizers or Combustibles)



Not Intrinsically Reactive, Flammable, or Combustible



Compatible Strong, Oxidizing Acids



Compatible Stable Explosives^{*}
(not including Oxidizing Explosives)



Flammables, Combustibles & Organic Solvents 



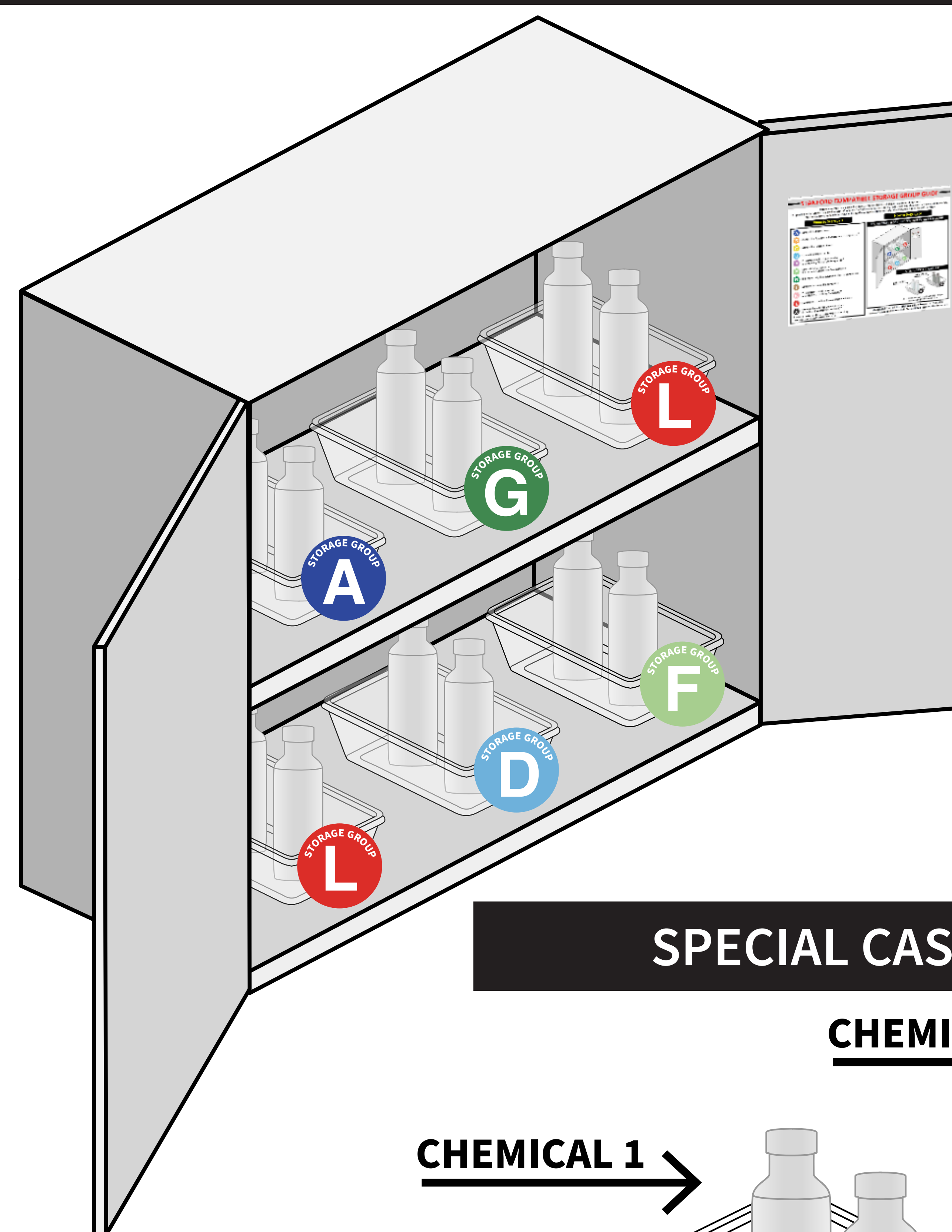
Incompatible with ALL Other Chemicals^{*}
(including other chemicals within X)

^{*} These materials are likely to require special handling & storage conditions. Use extreme caution.

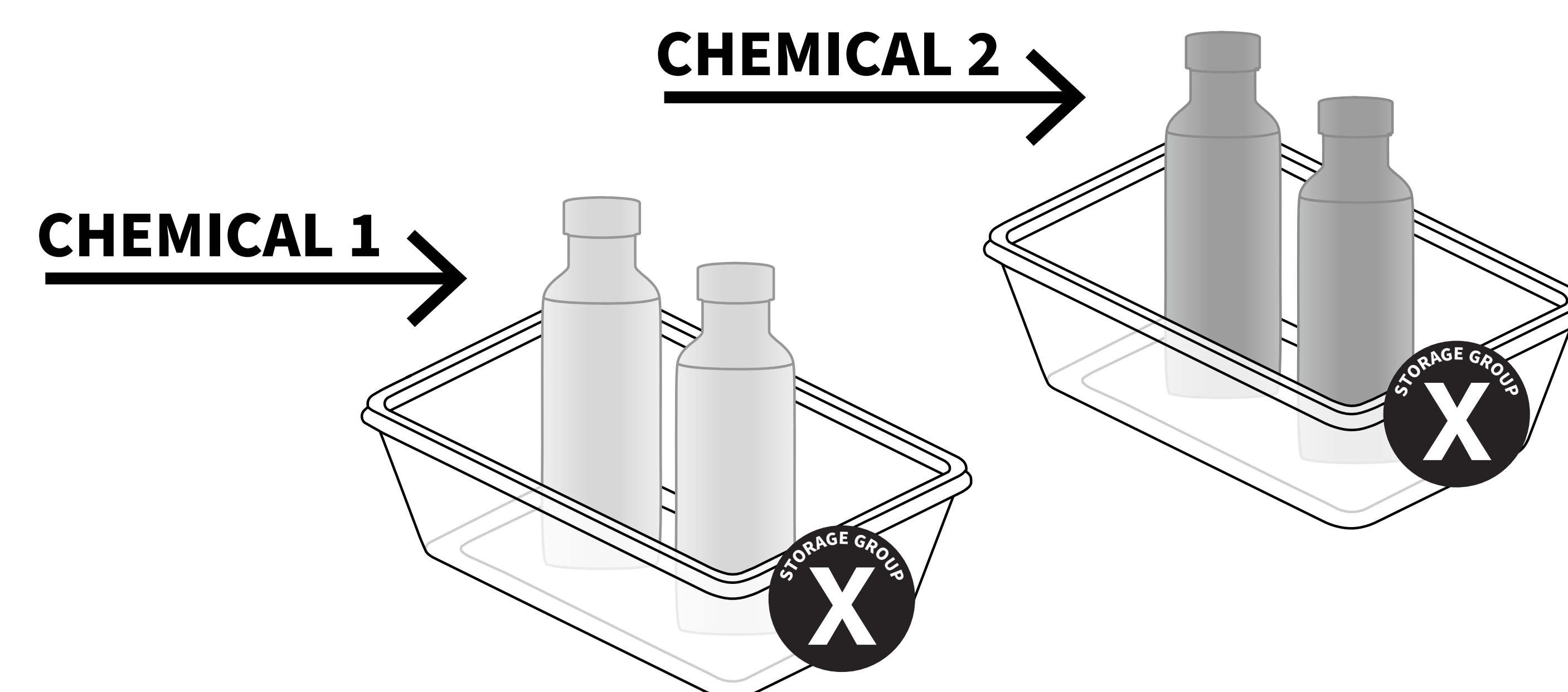
 These materials must be stored in a flammable cabinet.

How to Segregate

USE SEPARATE SECONDARY CONTAINERS FOR EACH GROUP



SPECIAL CASE FOR GROUP X



NOTE: Different chemicals within Storage Group X must be segregated from each other.

Questions? Contact the EH&S Lab Safety Program at 723-0448
Use ChemTracker to find a chemicals' Storage Group - stanford.chemtracker.org

Recommended Storage Groups for Common Chemicals

CHEMICAL	Group	Ethers	L	(K ₃ PO ₄)	
1-Butanol or 2-butanol	L	Ethidium bromide	G	Propionic acid	D
1-Propanol	L	Ethyl acetate	L	Propylene oxide	L
2-Mercaptoethanol	L	Ethylene glycol	L	Pump oil	L
Acetic acid, glacial (flammable)	D	Ficoll	G	Pyridine	A
Acetic anhydride (in THF or acetone: L)	X	Formaldehyde	L	SDS (Sodium dodecyl sulfate) (in solution: G)	L
Acetone	L	Formamide	L	Sigmacote	L
Acetonitrile	L	Formic Acid (≥85%)	D	Sodium acetate	G
Acetaldehyde	L	Glutaraldehyde	G	Sodium azide	X
Acrolein	X	Glycerol	L	Sodium bicarbonate	G
Acrylamide	G	Glycine	G	Sodium bisulfate	G
Agarose	G	Guanidine hydrochloride	G	Sodium bisulfite	G
Ammonium acetate	G	Guanidinium thiocyanate	C	Sodium borate	G
Ammonium chloride	G	Halothane, isoflurane	G	Sodium borohydride	B
Ammonium formate	G	HEPES	G	Sodium carbonate	G
Ammonium hydroxide	C	Hexanes	L	Sodium chlorate	E
Ammonium nitrate	E	Hydrochloric acid	F	Sodium chloride (NaCl)	G
Ammonium persulfate	E	Hydrogen peroxide, > 5%	E	Sodium citrate dihydrate	G
Ammonium sulfate	G	Hydrogen peroxide, < 5%	G	Sodium dichromate dihydrate	E
Ammonium sulfide	L	Imidazole	A	Sodium hydroxide (NaOH)	C
Benzene	L	Isobutyl alcohol	L	Sodium hypochlorite	E
Benzyl chloride	B	Isopentane	L	Sodium hypochlorite solution (i.e. bleach)	E
Benzoic acid	D	Isopropanol	L	Sodium phosphate	G
BIS/Bis-acrylamide	G	Lithium hydroxide	C	Sodium sulfide, anhydrous	B
BIS-TRIS	A	Magnesium chloride	G	Succinic acid	D
BIS-TRIS-HCl	G	Magnesium sulfate	G	Sucrose	G
Borax	G	Maleic acid	D	Sulfuric acid	I
Boric acid	G	Methanol	L	Tannic acid	G
Calcium chloride	G	N-Methyl-2-pyrrolidone	L	TEMED	A
Chloroform	G	N,N-Dimethylformamide	L	TES free acid	G
Chromic acid	I	Nitric acid	I	Tetracycline	G
Citric acid	D	p-Dioxane	L	Tetrahydrofuran	L
Coomassie Blue	G	Paraformaldehyde	L	Trichloroacetic acid	D
Dextrose	G	Perchloric acid	I	Trifluoroacetic acid	D
Dichloromethane	L	Periodic acid	I	Toluene	L
Diethylamine (flammable)	A	Permout	L	Triethanolamine	A
Diethyl pyrocarbonate (DEPC)	L	Phenol (solid)	G	TRIS	A
Dimethyl sulfoxide (DMSO)	L	Phenol (liquid, ≤ 89% phenol)	L	Triton X-100	G
Drierite	G	Phosphoric acid	F	Trizol	L
Econo-Safe, UniverSOL, BetaMax, CytoScint, Scintisafe, EcoLume, Ecoscint, Opti-fluor	L	Picric acid (any concentration)	X	TWEEN 20	G
EDTA (in solution: G)	D	Piperidine	A	Urea	G
Ethanol	L	PIPES, free acid	G	WD-40	L
Ethanolamine	A	Potassium acetate	G	Xylenes	L
		Potassium chloride	G	Zinc chloride	G
		Potassium cyanide	C		
		Potassium hydroxide (KOH)	C		
		Potassium phosphate	G		

Laboratory Chemicals & Waste Guidelines

Chemical Hazard Awareness

Hazard Communication and Globally Harmonized Labels



Corrosive: skin corrosives/burns, eye damage, corrosive to metals



Flammable: gas, liquid, solid, and aerosol



Acute Toxicity (fatal or toxic): known carcinogens or known mutagens; oral toxicity; toxic metals or pesticides; toxic to aquatic species



Aquatic Toxicity: effects of a material to cause injury or adverse effects to aquatic organisms in short-term or chronic exposures



Irritant (skin and eye): skin sensitizer, acute toxicity (harmful), narcotic effects, respiratory tract irritant, hazardous to ozone layer



Gases under Pressure: contained in a receptacle not less than 280 Pa at 20 °C or as a refrigerated liquid



Health Hazard: carcinogens, mutagens, reproductive toxins, respiratory sensitizers, target organ toxicity, aspiration toxicity



Explosives: self-reactives, organic peroxides

Hazardous materials are NEVER to be disposed of in the sink!



Note: This poster contains important regulatory information about hazardous materials and hazardous wastes that every laboratory worker is required to know. You may be asked to demonstrate your knowledge of these subjects by City, County or State inspectors.

revised: 5/2023

Handling Laboratory Wastes

A laboratory chemical becomes a waste when you no longer intend to use or re-use the chemical.

Laboratory wastes may be accumulated for up to a maximum of 9 months in the lab.



- A waste tag must be created at the **START** of waste accumulation
- An online waste tag can be created and printed for your use at WasteTag.stanford.edu
- Submit a pickup request as soon as the container is full or 8 months after the accumulation start date, whichever comes first



Hazardous Waste Categories



Corrosive



Flammable



Oxidizer



Air or Water
Reactive

Chemical wastes that are not clearly in one of the first four categories, and are not listed on the Stanford non-hazardous waste list NonHazardousWaste.stanford.edu should be considered toxic



Toxic

Waste containers must meet the following criteria:



- Good condition with no leaks or cracks
- Kept closed when not actively adding waste to the container
- Properly segregated from incompatible materials
- Always stored within secondary containment
- Affixed with a fully completed hazardous waste label

All chemicals not specifically listed on the Stanford University "Non-Hazardous Waste List" must be managed as a hazardous waste.



- See: NonHazardousWaste.stanford.edu for a list of non-hazardous wastes.
- If your material is on the non-hazardous waste list, follow the instructions for alternative disposal.

Spill Response

Call **SoM Security** for EHS cleanup assistance if:



- You are not knowledgeable of the material and its hazards, or
- The spill is not contained in a hood or on a lab bench, or
- The spill may result in an environmental impact by entering a sink or floor drain, or by contaminating soil, or by producing hazardous vapor emissions, or
- You cannot complete the cleanup with your materials on hand
 - Report spills greater than 30 ml by calling (650) 725-9999, EHS documents large spills

You do not need to call for spill assistance if:



- You are knowledgeable of the hazards of the material
- The chemical spill is less than 30 ml
- It takes less than 15 minutes to clean up
- You can clean it up using available spill response materials and personal protective equipment
 - All contaminated spill control cleanup materials and personal protective equipment must be managed as hazardous waste
 - You must document self-cleanup

Accident Response

Emergency Response for Accidents Involving Hazardous Materials

If the accident is a fire, explosion, or health-threatening:

- Dial **911**, then dial **SoM Security** for emergency assistance
- Follow your department emergency procedures
- Evacuate the area or building

In case of eye or skin contact with hazardous chemicals:

- Immediately flush the affected area with water for 15 minutes
- Contact the Stanford Occupational Health Clinic at (650) 725-5308 during business hours (For all other times, dial **911**)
- Notify your supervisor
- Have a co-worker call **SoM Security** for emergency assistance
- Complete a SU-17 (incidentreport.stanford.edu) and submit the form to Risk Management
- If the injury meets the serious injury/illness criteria (requires inpatient hospitalization, loss of limb, or permanent disfigurement), call (650) 725-9999 to report



SoM Security (650) 723-7222
SoM Work Flow Desk (650) 721-2146
medfacilities.stanford.edu

Stanford School of Medicine

Contact Notification Sheet

Room: _____

Building (QB): _____

Department/Division/Work Gp (OrgCode): _____

In Case of Emergency, Notify the Following Individuals

Room Level:

Role	Name (SUNet)	Room	Primary Phone	Alternate Phone
Chemical Owner/PI				
Lab Contact 1				
<i>Lab Contact 2</i>				
<i>Lab Contact 3</i>				
<i>Lab Contact 4</i>				

Department/Organizational Level:

Role	Name (SUNet)	Room	Primary Phone	Alternate Phone
DFA				
Safety Contact				
Emerg Coordinator				
<i>Addnl Dept Contact</i>				
<i>Addnl Dept Contact</i>				
<i>Addnl Dept Contact</i>				

Building/Facility Level:

Role	Name (SUNet)	URL/Email	Phone
Facility Operations	SOM Workflow Control Desk	medfacilities.stanford.edu	650-721-2146
SOM Safety Contact			
SOM Emerg Contact	Monica Dibble	mdibble@stanford.edu	650-723-9600
<i>Building Manager</i>			

FOR MEDICAL EMERGENCIES

CALL: 9-911

Also notify Security:

Off Campus: (650) 721-0762 or

Main Campus: (650) 723-7222

Review (Date/Initial)

INSTRUCTIONS FOR COMPLETING THE STANFORD SOM CONTACT NOTIFICATION SHEET

1. Labs Departments are responsible for maintaining current contact information for Room and Department/Organizational Level Contacts and posting this information inside the Life Safety Box.
2. Contact the Safety and Emergency Management (S&EM) representative for your building for assistance in completing the form (<http://med.stanford.edu/medfacilities/general-safety/safety--and-emergency-management-contacts.html>)
3. Review the information at least annually or when there is a change in Room, Department/Organization or Building/Facility Contact; Place the completed form inside the life safety box posted at the entrance.

SECTION 1 – LOCATION

The information in this section is obtained from ChemTracker and iSpace for rooms reporting a chemical inventory.

- Room:
- Building (QB):
- Department/Division/Work Group/(OrgCode):

SECTION 2 – ROOM LEVEL

The name of the Chemical Owner/PI is from ChemTracker. Please provide the SUNetID, primary work location, along with primary and alternate phone numbers (include area code and full 7-digit phone number) for the Chemical Owner; identify at least one contact present in the laboratory or work area and provide similar contact information.

- Chemical Owner/PI
- Lab Contact(s) – representative designated by PI who is knowledgeable on lab research procedures, practices or equipment in the room
- Optional contacts can also be listed on the form if appropriate including Chemical User (for updating chemical inventory information), Husbandry Contact, Husbandry Supervisor, Operations Manager, etc..

SECTION 3 – DEPARTMENT/ORGANIZATIONAL LEVEL

Includes DFA, Safety and Emergency Coordinator designated by department to coordinate safety, compliance and emergency management for the department or organizational unit.

- DFA
- Safety Contact (individual(s) representing department, division, department operations or programs and responsible for coordinating safety or compliance related issues)
- Department Emergency Coordinator is the primary contact from the department and is responsible for coordinating emergency and recovery planning.
- Additional department or organizational level contacts to include if applicable include: Program Manager, Director, Division Manager, Operations Manager, Department Facilities Coordinator/Director

SECTION 4 – BUILDING/FACILITY LEVEL

- Facility Operations – Contact information for the OFPM Workflow Control Desk; use for reporting housekeeping or other facility-related issues or services.
- SOM Safety Contact - S&EM representative for your building
- SOM Emergency Contact
- Building Manager- There are only a few SOM Buildings that have a designated Building Manager; check with your S&EM Contact for details.

SECTION 5 – HEALTH-THREATENING EMERGENCY

Contact information for reporting a medical emergency

- Call: 9-911
- Also notify Security: Off-Campus (650) 721-0762 or Main Campus: (650) 723-7222

SECTION 6 – REVIEW

Review the information at least annually; Print out an updated form and Date and Initial the form.

Chemical Storage Map

Building:	Room:	Date:	
		 Life Safety Box	
		 Fire Extinguisher	
		 Shower	
		 Eyewash	
		 Shower & Eyewash	
		 Free-Standing Cabinet	
		 Fume Hood	
		 Natural Gas Outlet	
		 Gas Cylinder	
		 Floor-Standing Equipment	
Hazards Key		 Refrigerator and/or Freezer	
Hazard #	Description		Threshold Quantity
1	Explosive	Forbidden	 Door
2.1	Flammable Gases	1 cu ft	
2.2	Non Toxic/Flam Gases	1 cu ft	
2.3	Toxic gases	1 cu ft	
3	Flammable Liquid	5 gals or 40 lbs	
4.1	Flammable Solid	40 lbs	
4.3	Water Reactive Flammable	1 gal or 8 lbs	
5	Oxidizer	1 gal or 8 lbs	
6.1	Toxic Solid or Liquid	1 gal or 8 lbs	
7	Radioactive	Level B Storage or Radiation Producing Machines	
8	Corrosive	5 gals or 40 lbs	
9	Other Miscellaneous Hazards	Above as appropriate	
BL2/3	Biohazards	Any material containing Biosafety level 2 or 3 agents	

INSTRUCTIONS FOR COMPLETING THE ROOM HAZARD MAP FOR THE LIFE SAFETY BOX

1. General

Complete the top of the map with your building name, room number, and the date.

2. Equipment

- a. Draw an outline of the room and fixtures including:
 - Doors
 - Benches and cabinets
 - Floor-standing equipment
- b. Indicate the location of safety equipment using the symbols shown on the right side of the Map template.
- c. Label refrigerators, floor-standing cabinets, fume hoods, and gas cylinders using the symbols shown on the right side of the map template.
- d. Label natural gas outlets with the symbol shown on the map.

3. Chemical Hazards

- a. Remove the "Guide to Hazard Quantities" and the "Hazardous Materials Inventory" from the Life Safety Box.
- b. Compare the quantities of each Hazard Category given on the "Guide to Hazard Quantities" in the LSB box Printouts to the threshold in the "Hazards Key" on the map template.
- c. If the quantity stored exceeds a corresponding threshold, mark the main storage areas for the Hazard Category with the corresponding hazard number on the map. If necessary, check the detailed "Hazardous Materials Inventory" to identify the specific compounds that fall within the Hazard category and then determine where those compounds are stored.

(For example, your "Guide to Hazard Quantities" indicates that you have a total of 3 gallons of oxidizers and 1 pound of moderate poisons. The amount of oxidizer exceeds the threshold of 1 gallon given on the Hazards Key and since you store most of this material in two locations, you mark both of these locations on the map with a number 5. A container of dilute oxidizer solution is stored on the bench top and because this is a minimal amount, you do not mark this location on the map with the number 5. The amount of moderate poison does not exceed the threshold of 8 pounds and thus you do not mark the map with hazard number 6.1.)

- d. Stop after you reach Hazard Category 9.

4. Other Hazards

If applicable, use the number or letters annotated on the Hazards Key to indicate the location of the following hazards:

- Biosafety Level 2 or 3 materials storage
- Radiological materials storage
- Radiation producing machines

INCIDENT REPORTING

Exposure Response

for biological, chemical, or radiological exposures

CALL 911 FOR ANY HEALTH-THREATENING EMERGENCY, THEN FOLLOW THESE STEPS

Perform First Aid	
Needlestick, Puncture or Sharps Injury, or Animal Bite/Scratch	Wash for 15 minutes with warm water and soap. Radioactive exposure: Squeeze wound to encourage bleeding, then rinse generously using a mild soap.
Eye Exposure	Use emergency eyewash to flush eyes for 15 minutes while holding eyes open. If wearing contact lenses, remove them as soon as practical, but do not delay rinsing while waiting for contact lens removal.
Skin Exposure	Remove contaminated clothing and wash skin with soap and water for 15 minutes. WHEN IN DOUBT, USE THE SAFETY SHOWER
Inhalation	Move out of the contaminated area and seek fresh air. Radioactive exposure: Blow nose into clean tissue and survey for contamination. Contact Health Physics. 650-723-3201
Get Medical Help	
Life- or Health- Threatening Exposure	Call 911 or proceed to the Stanford Hospital Emergency Department (1199 Welch Road, Palo Alto). Then, call the EH&S incident hotline at 650-725-9999 (or have someone call for you).
Urgent Exposure	Call the EH&S incident hotline at 650-725-9999.
Non-Urgent Exposure	Contact the Occupational Health Center for medical consultation at 650-725-5308, Monday – Friday from 8 AM – 4 PM. If the Occupational Health Center is closed and you feel you need immediate medical attention, proceed to the Stanford Hospital Emergency Department. If you do not feel immediate assistance is necessary, call the Occupational Health Center the following day.
Essential Information to Give Medical Professionals	Chemical: chemical name, quantity, concentration; bring Safety Data Sheet (SDS) with you Biological: pathogen name, quantity, concentration, any known antibiotic or antiviral resistances Radioactive: isotope, isotope activity All: route of exposure, duration of exposure, time since exposure
For All Exposures	Notify your supervisor. Secure the area before leaving.
Report the Incident	
All Incidents and Near Misses	Submit a report via the incident report form (SU-17) at su17.stanford.edu .

Serious Injury/Illness Reporting Procedures

<https://ehs.stanford.edu/topic/occupational-injury-illness/serious-injuryillness-reporting-procedures>

What is a serious injury/illness?

Per Cal/OSHA regulation, employers shall report any work-related serious injury, illness, or death to Cal/OSHA as soon as practically possible, but not longer than eight hours after the incident. Cal/OSHA defines an injury or illness as serious if it:

- Requires inpatient hospitalization for reasons other than medical observation or diagnostic testing
- An employee suffers a loss of any limb of the body
- An employee suffers any serious degree of permanent disfigurement

Stanford reporting procedure

In the event of serious injury, illness, or death:

1. Call the EH&S Emergency Hotline at (650) 725-9999 immediately. (During non-business hours, this number will reach a University operator, who will collect incident information and contact EH&S.)
2. The Occupational Health and Safety (OHS) Program staff will take the facts about the incident and the status of the injured employee(s) from the supervisor and/ or the employee(s).
3. The OHS Program staff will report the incident to the Cal/OSHA district office.
4. The supervisor must complete the requisite forms (refer to the table below or contact Risk Management at (650) 723-7400).
5. The supervisor must follow up on the incident and document corrective actions (using the [Incident Investigation Report](#)). EH&S will review all forms and contact supervisors as needed for follow up. The goal of incident investigations is to prevent incidents from recurring, not to assign culpability.

Who calls Cal/OSHA?

- EH&S is the designated University authority to report serious incidents to Cal/OSHA. The EH&S emergency reporting number (5-9999) is monitored 24 hours a day.
- **Supervisors and co-workers must not contact Cal/OSHA directly to report serious incidents.**

What are the time limits for reporting?

- EH&S shall report all serious incidents, injuries, and illnesses to the nearest District Office of Cal/OSHA as soon as practically possible, but not longer than eight hours after the incident.
- Failure to immediately report serious incidents and injuries to EH&S can result in Cal/OSHA citations and fines. Note that regulatory violations are paid by the operating unit responsible for the citations.

Additional information on incident reporting at Stanford University

- See the table below for Stanford University work-related incident/injury/illness reporting forms.
- If necessary, contact EH&S at (650) 723-0448 or [visit the IIPP website](#).

- If necessary, contact the [Office of Risk Management](#) at (650) 723-7400.
- Review the Stanford University Administrative Guide:
 - Emergency/Accident Procedures (25.2)
 - Accident and Incident Reporting (25.6)

Stanford University work-related incident/injury/illness report forms

Supervisors are responsible for reporting all work-related incidents, injuries, and illnesses. See below for the reporting required.

Incident/injury/exposure report forms

Form	When To Use	Instructions
SU-17 – Incident Investigation Report	Any incident involving a Stanford University employee, student, visitor, and/or contractor (note: use the SU-17B for non-University employees)	Submit to Risk Management within 24 hours
State Form DWC-1 (workers compensation)	Provide to employees when they visit a doctor due to a work-related injury/illness	Complete and provide to employee within 24 hours of incident or report of incident
SU-16 – Workers’ Compensation Lost Workdays Report	Provide to employees and Risk Management when they visit a doctor due to a work-related injury/illness.	Supervisors/HR Managers must complete and provide to the employee within 24 hours of incident or report of incident.
Cal/OSHA 5020- Employer’s Report of Occupational Injury or Illness	Supervisors/ HR Managers must complete for employees when treatment is required by a physician/provider in a medical facility.	Submit to Risk Management as soon as possible, but within 5 days

SU-17 INCIDENT REPORTING IS ESSENTIAL TO A HEALTHY SAFETY CULTURE

1 WHAT TO REPORT

- Injuries
- Close Calls
- Exposures
- Fires
- HazMat
- Spills

2 HOW TO REPORT

Go to **su17.stanford.edu**

3 WHEN TO REPORT

As soon as possible! And within the same day.

SERIOUS INJURIES

For incidents resulting in death, amputation, loss of an eye, disfigurement, and/or in-patient hospitalization, ***immediately call 650-725-9999***

4 WHAT HAPPENS WHEN YOU REPORT

After you report, safety professionals will:

- Follow up and gather information
- Perform necessary analysis
- Assign corrective actions (if applicable)
- Promote learning and communications

Taking these steps will help make Stanford a safer place to work, study, and conduct research.



Learn more or submit a report at
SU17.STANFORD.EDU



STANFORD UNIVERSITY

OCCUPATIONAL HEALTH CENTER

MEDICAL CARE IN EMERGENCIES & AFTER HOURS

For immediate, life-threatening
emergencies, call 911
(or 286 in Medical Center)

and/or go to the

Stanford Hospital
Emergency Department
1199 Welch Road
Palo Alto, CA 94304
(650) 723-5111

CONTACT US

Phone: (650) 725-5308
Email: stanfordohc@stanford.edu
Website: suohc.stanford.edu

VISIT US

Environmental Health & Safety
484 Oak Road (2nd Floor)
Stanford CA, 94305-8007

HOURS

8:00 am – 4:00 pm
Monday – Friday

On-site services for
Stanford University
faculty, staff, and postdocs'
work-related:

- Medical surveillance
- Immunizations
- Injuries & Illnesses
- Travel Consultations



REPORTING EMPLOYEE WORK-RELATED ACCIDENTS, INCIDENTS & EXPOSURES

1. Inform Supervisor
2. Call SU Occupational Health Center (SUOHC) at 650-725-5308, if event involves a work-related injury or illness
3. Complete an SU-17, at <https://SU17.stanford.edu>

MEDICAL STAFF

Rich Wittman, MD, MPH
Raj Puri, MD, MPH
Minal Moharir, MD
Clare McNamara, PA

ONSITE SERVICES

MEDICAL SURVEILLANCE

Medical Surveillance is the process of evaluating workers' health as it relates to their potential occupational exposures to hazardous agents, it includes:

- Tuberculosis screening
- Vision Exams
- Respirator Use Clearances
- Hearing Tests
- Focused Physical Exams
- Urinalysis
- Blood Tests

IMMUNIZATIONS

- MMR (Measles, Mumps, Rubella)
- Varicella (Chickenpox)
- Hepatitis B
- Tetanus and Pertussis
- Rabies
- Others as indicated for work-related travel or exposures

EMPLOYEE WORK RELATED INJURY AND ILLNESS CARE

- Diagnostic Testing
- Medications
- Exercise / Stretching Programs
- Referral to Physical Therapy
- Work Status Reports
- Medical Treatment for:
 - First Aid – Cuts, Abrasions
 - Sprains / Strains (e.g., back, knee, wrist)
 - Repetitive Stress Injuries
 - Bloodborne Pathogen Exposure (e.g., needle sticks, blood splash)
 - Animal Bites
 - Non-Human Primate Exposures
 - Laser Exposures
 - Radiation Exposures

EMERGENCY PREPAREDNESS FOR STUDENTS, FACULTY, STAFF, & VISITORS

Emergencies or disasters can happen any time, and they usually occur without warning. When an emergency (such as a fire, earthquake, hazardous materials release or utilities outage) strikes at Stanford, our safety and prompt recovery will depend on existing levels of preparedness and a thoughtful response among students, faculty, staff and visitors.

HOW TO PREVENT CAMPUS EMERGENCIES & PROTECT YOUR OWN SAFETY

FIRE PREVENTION

- ☐ Note the location of alarms and extinguishers. Know how to use them
- ☐ Leave fire doors closed at all times
- ☐ Clear obstructed corridors, aisles and room exits
- ☐ Use only grounded electrical plugs. Limit use of extension cords and multiple outlets
- ☐ Do not use mechanical rooms or utility rooms for storage
- ☐ Do not smoke in University buildings

EARTHQUAKE PREPAREDNESS

- ☐ Know how and where to take cover during a quake
- ☐ Anchor furniture over 42". Do not stack furniture
- ☐ Move tall bookcases away from exits. Do not use them as room dividers
- ☐ Anchor computers, equipment, and display cases
- ☐ Store heavy items at floor level
- ☐ Back up critical data. Keep duplicates off-site

LABORATORY SAFETY

- ☐ Maintain a clean work environment
- ☐ Post lab safety work rules, train all personnel
- ☐ Inventory and label chemicals. Segregate incompatibles
- ☐ Keep flammables in flammable storage cabinets
- ☐ Do not block access to eyewashes, showers, or fire extinguishers
- ☐ Keep copies of Safety Data Sheets. See <https://ehs.stanford.edu/services/safety-data-sheet-database/>
- ☐ Back up cultures and data off-site
- ☐ Investigate emergency power options
- ☐ Install seismic restraints on chemical storage shelves
- ☐ Anchor equipment and furniture. Latch cabinet doors. Avoid high storage of heavy items
- ☐ Chain compressed gas cylinders at 1/3 and 2/3 points
- ☐ Do not store hazardous materials on mobile carts
- ☐ Dispose of chemical waste properly. See: <https://ehs.stanford.edu/services/chemical-waste-pickup>

EMERGENCY KIT CONTENTS

FOR WORK

- ☐ Flashlight, portable radio and batteries
- ☐ First aid supplies and instructions
- ☐ Sturdy closed shoes
- ☐ Extra glasses, critical medications
- ☐ Container of water and non-perishable snack
- ☐ Out-of-state telephone contact, coins or calling card

IN THE CAR

- ☐ Flashlight, portable radio and batteries
- ☐ First aid supplies and instructions
- ☐ Sturdy closed shoes, jacket, socks, umbrella
- ☐ Extra glasses
- ☐ Container of water and non-perishable snack
- ☐ Auto fire extinguisher
- ☐ Tools: jumper cables, crowbar, rope
- ☐ Flares, light sticks, weatherproof matches
- ☐ Out-of-state telephone contact, coins or calling card

AT HOME

- ☐ Flashlight, portable radio and batteries
- ☐ First aid supplies and instructions
- ☐ Sturdy shoes (also keep shoes & a flashlight under the bed)
- ☐ Extra glasses, medications, sanitation supplies
- ☐ Fire extinguisher
- ☐ Bottled water (1 gallon/person or pet/day)
- ☐ Emergency food for at least 3 days (people and pets)
- ☐ Cooking supplies: camp stove/barbecue, manual can opener, paper plates, utensils
- ☐ "Space" blankets, extra clothing, waterproof matches
- ☐ Special items for pets, infants, elderly or disabled household members
- ☐ Tools: adjustable wrench, crowbar, axe, shovel, broom, heavy work gloves
- ☐ Staple gun, heavy tape, plastic sheeting for windows
- ☐ Out-of-state telephone contact, address book
- ☐ Cash (including small bills and coins)
- ☐ Important documents (insurance, bank, real estate)

EMERGENCY COMMUNICATIONS

Depending on what is appropriate for the situation, information will be available through the following:

- ☐ Notices posted on the campus emergency website (<http://emergency.stanford.edu>) or emergency information hotlines (650-725-5555, 844-253-7878)
- ☐ Messages broadcast on KZSU 90.1FM or our outdoor warning system
- ☐ Alerts communicated through our AlertSU mass notification tool, via phone, email, and text messaging.
- ☐ To keep your AlertSU contact information up to date, go to <http://stanfordyou.stanford.edu> (faculty/staff) or <http://axess.stanford.edu> (students).

BUT DO YOU KNOW WHAT TO DO WHEN SOMETHING HAPPENS?

WHAT TO DO WHEN AN EMERGENCY OCCURS

ACCIDENT

- ☐ Call 9-911 (911 from non-SU phones, 286 from Med Center)
- ☐ Administer first aid if you are trained to do so
- ☐ Do not attempt to move seriously injured persons

FIRE

- ☐ Call 9-911 (911 from non-SU phones, 286 from Med Center)
- ☐ Activate nearest alarm
- ☐ Feel doors for heat. If cool, exit carefully. If hot, do not open the door. Stay where you are
- ☐ If you see smoke, crouch near the floor as you exit
- ☐ If you see fire, confine it by closing doors and windows
- ☐ Use extinguishers on small fires only if it is safe to do so
 - PULL the Pin, AIM at the base of the fire
 - SQUEEZE the nozzle, SWEEP back and forth
- ☐ Never use an elevator during a fire evacuation
- ☐ Go outdoors to the Emergency Assembly Point (EAP)

HAZARDOUS MATERIALS SPILL

MINOR release in the lab

- ☐ Follow eyewash, rinse, or shower procedures
- ☐ Vacate persons in immediate area if necessary
- ☐ Clean spill if you have suitable training
- ☐ Wear protective equipment
- ☐ Use appropriate kit to contain, neutralize and absorb spill
- ☐ Collect, containerize and label waste. See: <https://ehs.stanford.edu/services/chemical-waste-pickup>

MAJOR release in the department

- ☐ Call 650-725-9999 (Med Center, call 286) to report the spill.
- ☐ Assist injured persons
- ☐ Isolate contaminated persons
- ☐ Close doors or control access to spill site
- ☐ Alert Supervisor or Department Chair
- ☐ Follow evacuation instructions precisely

EARTHQUAKE

- ☐ Do not run or panic
- ☐ Take cover immediately
 - ☐ Under a desk, table or chair
 - ☐ Between seating rows in classrooms
 - ☐ Against a corridor wall
 - ☐ Outdoors – away from falling hazards
- ☐ Expect aftershocks

MAJOR QUAKE (violent motion)

- ☐ Restore calm. Assist others
- ☐ Report injuries or fires to 9-911(911 from non-SU phones, 286 from Med Center)
- ☐ Report damage to department head
- ☐ Evacuate carefully
- ☐ Take emergency supplies
- ☐ DO NOT USE ELEVATORS
- ☐ Go outdoors to the Emergency Assembly Point (EAP)
- ☐ Do not enter buildings until they are examined
- ☐ Await instructions, be patient, help others

MINOR SHAKE (brief rolling motion)

- ☐ Restore calm. Examine your area for damage
- ☐ Report damage/hazardous releases



Stanford University Environmental Health & Safety

ESF - 480 Oak Road

Stanford, CA 94305

Phone: 650-723-0448

Fax: 650-725-3468

<http://ehs.stanford.edu>